

Chaos in the phase space formulation of quantum mechanics

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DECLARATION

I hereby declare that the investigation presented in the thesis has been carried out by me. The work is original and has not been submitted earlier as a whole or in part for a degree/diploma at this or any other Institution/University.

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Abstract

The Wigner function reformulates quantum mechanics into a quasi-phase space theory where averages of observables and time evolution is similar to classical probability and Liouvillian dynamics respectively. This thesis dives into the structural framework of the Wigner function to study quantum chaos. Quantum chaos does not have a universal definition due to unitary time evolution, discreteness of the energy spectrum, and many other reasons. Since the Wigner function is a phase space formulation, it offers a natural theoretical tool to study chaos in quantum systems. The negativity of the Wigner function, i.e., the total volume of its negative part was used as the primary diagnostic of chaos in this thesis.

Several quantum maps exist which have been extensively studied. The first ever quantized map was the quantum standard map on the torus. In simple terms, this system represents a free particle that is perturbed by a delta-function impulse at regular intervals of time. Floquet theory was employed to create the unitary operator that controls the dynamics of the system. An immense collection of literature exists on this system but very few have studied its Wigner function negativity for signatures of chaos. In this thesis, the negativity of the Wigner function was studied as a function of time and the various parameters of the quantum standard map on a torus. The analysis was mostly done numerically and then understood in analytical terms. Time evolution of the Wigner function offered the most comprehensive measure of chaos in the system. Extensive analysis is done for the time evolution of the negativity which was obtained numerically and understood in analytical terms.

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Chapter 1

Introduction

The Heisenberg uncertainty principle states that non-commuting observables cannot be measured to infinite precision simultaneously. This prevents the formulation of quantum mechanics in terms of a phase space picture analogous to classical mechanics where the evolution of a system describes a trajectory in phase space. Instead of a phase space, quantum mechanics is described in terms of the Hilbert space. The evolution of a quantum system is understood as the rotation of an infinite dimensional unit vector along the surface of an infinite dimensional hypersphere in Hilbert space. Feynman introduced a different formulation which generalized the principle of least action from classical mechanics to quantum mechanics. In this formulation the evolution of a quantum system takes place over multiple possible paths from an initial to final state. The contribution from each path is posited as a complex exponential whose phase is the classical action and the wavefunction corresponds to the total contribution of all possible paths. This describes the path integral formulation of quantum mechanics [1].

Despite the limitations placed by the uncertainty principle, there exist multiple different versions of a phase space formulation which recasts quantum mechanics on the (q, p) phase space while accounting for their non-commutativity. The Wigner function provides one such phase space formulation. It has an inherent philosophical advantage over the Hilbert space formulation. In the latter, probabilities are an *ad hoc* interpretation of the wavefunction as postulated by Born's rule. The Wigner function being a quasi-phase space formulation has probability built into its theoretical framework [2]. However, it would be wrong to say that the phase space structure of the Wigner function provides a complete description of probabilities since it can have negative values. More will be said about this in later sections but an important thing to note is that the negative values play a big part in the analysis performed in this thesis. The quasi-phase space structure of the Wigner function offers a way to explore manifestations of chaos

in quantum systems. Quantum chaos is a field of study that studies signatures of chaotic behaviour in quantum systems which are known to be classically chaotic. In classical systems, phase space mixing, positive Lyapunov exponent, and an extreme sensitivity to initial conditions, are some of the known characteristics of chaos. While there are certain analogous quantities, such as the quantum Lyapunov exponent, they do not form a rigorous definition of quantum chaos. Sensitivity to initial conditions, a hallmark of classical chaos, does not exist in quantum mechanics since the Schrödinger equation is linear and the evolution of a system is given unitary. Unitarity of time evolution guarantees that the probabilities remain bounded by 1 and the linearity of the Schrödinger equation means that a small change in the initial condition will not be amplified by time evolution. It is important to note that the linearity of the Schrödinger equation does not imply that chaotic behaviour cannot manifest in quantum systems. Liouville's equation, that describes the evolution of a classical system, is also a linear system. The signatures of quantum chaos lie elsewhere. One of the most notable features of quantum chaos is manifest in the spectral characteristics of the eigenstates of the Hamiltonian of a quantum system which exhibits classical chaos. It was shown by [3], [4] and many others that the statistical properties of the energy eigenvalues of the quantum version of a classically chaotic system follow the statistics of Random Matrix Theory.

The most well known systems that are studied to understand quantum chaos are quantum maps. These quantum maps are area preserving maps that are usually defined on a torus, i.e., the allowed values of q and p are both bounded between 0 and 1. Such quantum maps provide very simple systems that can be numerically studied while also allowing for analytical treatments. Some examples of such maps are the Chirikov map, Arnold's cat map, Baker's map, and many more. Studying the Wigner function of these systems provides a great method of probing manifestations of chaos in the phase space formulation of quantum mechanics. Extensive study of quantum maps would allow correspondence of obtained results with existing results. Thus, quantum maps are the best suited systems to use the Wigner function to probe chaos.

The thesis is structured as follows. Chapter 2 introduces the Wigner function and discusses its properties. The equivalence to the Hilbert space picture is also discussed here. Following this the finite dimensional formulation of the Wigner function provided by Wootters [5] is discussed in Chapter 3. In Chapter 4 the quantum standard map on a torus is introduced and discussed briefly. This system is the quantization of the Chirikov map with both q and p periodic in $(0, 1)$. The general properties of a quantum map on a torus

¹ are discussed here. Floquet theory is introduced to explain the dynamics of the quantum map and the Wigner function of eigenstates of the Floquet operator are also studied. The central analysis of the thesis is contained in Chapter 5 where the negative part of the Wigner function is used to study signatures of chaos in the quantum standard map on a torus are studied. The Wigner function negativity is introduced and its variation with the different parameters of the map are analyzed.

The use of the Wigner function to study quantum chaos is apt since it provides a quasi-phase space formulation. It is also very handy to probe the quantum-classical correspondence. Various properties of the Wigner function negativity are studied for the quantum standard map on a torus. Its behaviour in various parameter regimes are studied to understand manifestations of chaos in this specific quantum map.

¹Also known as the quantum kicked rotor.

Chapter 2

The Wigner Function

The formulation of classical mechanics in terms of Hamilton's equations of motions allows representing the evolution of a system in a well defined phase space of position and momentum in classical mechanics. However, the uncertainty principle prevents an analogous phase space picture. Efforts have still been made to provide a phase space understanding of quantum mechanics. The Wigner function provides one such representation. It was proposed by Eugene Wigner [6] as an extension of the Maxwell-Boltzmann distribution to quantum mechanics by finding an expansion in terms of $\hbar$. The complete understanding of quantum mechanics in terms of the Wigner function was provided by José Moyal [2]. The essential idea behind using the Wigner function relies on the proper definition of functions of non-commuting variables (q and p being the most common). Ordering of functions of non-commuting variables is expressed through Weyl ordering [7]. Moyal connected the expectation values of Weyl-ordered operators to the Wigner function and also provided an expression for the time evolution of the Wigner function thus establishing a proper theoretical framework of quantum mechanics in phase space.

The Wigner function is defined as [8][9][10]:

$$W(q, p) = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dy e^{ipy/\hbar} \langle q - y/2 | \hat{\rho} | q + y/2 \rangle \quad (2.1)$$

It is mathematically a Fourier transform of the shifted elements of the density matrix. For pure states $\langle q - y/2 | \hat{\rho} | q + y/2 \rangle$ reduces to $\psi^*(q + y/2)\psi(q - y/2)$. Eq. (2.1) can easily be generalized to N dimensions by taking the integral over $3N$ position and momentum vectors with the factor outside becoming $1/(2\pi\hbar)^{3N}$.

The Wigner function can also be expressed by taking the density matrix in the momentum basis by

performing a Fourier transform of the position basis wavefunction.

$$W(q, p) = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dk e^{-ikx/\hbar} \langle p - k/2 | \hat{\rho} | p + y/2 \rangle \quad (2.2)$$

The description is quite convenient because an integral over either position or momentum gives back the momentum space and position space probabilities respectively.

$$\int_{-\infty}^{\infty} W(q, p) dq = \langle p | \hat{\rho} | p \rangle \quad (2.3)$$

$$\int_{-\infty}^{\infty} W(q, p) dp = \langle q | \hat{\rho} | q \rangle \quad (2.4)$$

The average of an observable is the given by a weighted sum over the phase space. The form is the same as that of for any other variable with a given phase space distribution function. As a first assumption the average of an observable, $\hat{A}$, is

$$\langle A \rangle = \int dq dp A(q, p) W(q, p) \quad (2.5)$$

Where $A(q, p)$ is the phase space function corresponding to the quantum operator $\hat{A}$ and $W(q, p)$ is a Wigner function corresponding to some state $\hat{\rho}$. It is straightforward to prove that (2.5) is the correct expression for the average. Before proceeding with the proof it is important to understand the exact mapping between operators and phase space functions.

The validity of this formula depends on the correspondence of $\hat{A} \rightarrow A(q, p)$. There is a well-defined correspondence principle between the phase space function and the operator known as Weyl's correspondence (or transform). It was given by Weyl in his 1927 paper originally published in German. The transform assumes that the phase space function $A(q, p)$ has a Fourier inverse, $\alpha(\sigma, \tau)$

$$\alpha(\sigma, \tau) = \int dq dp A(q, p) e^{-i(\sigma q + \tau p)}$$

Given this then the operator $\hat{A}$ is

$$\hat{A} = \frac{1}{(2\pi\hbar)^2} \int d\sigma d\tau dq dp A(q, p) e^{i[\sigma(\hat{q}-q) + \tau(\hat{p}-p)]} \quad (2.6)$$

It should be noted that σ has dimensions of momentum while τ has dimensions of space. The validity of this transformation can be checked by checking whether it gives the correct ordered operator form of a

phase space function like xp . The inverse Weyl-transform can be obtained from the above representation.

$$A(q, p) = \int dy e^{ipy/\hbar} \langle q - y/2 | \hat{A} | q + y/2 \rangle \quad (2.7)$$

Given (2.7) we can see that the function corresponding to the density matrix, $\hat{\rho}$ is just the Wigner function divided by a factor of $2\pi\hbar$. This comes in handy to explain the negativity of the Wigner function which is one of its most striking properties. Consider the following relation

$$\text{Tr}\{\hat{\rho}_a \hat{\rho}_b\} = \int dq \langle q | \psi_a \rangle \langle \psi_a | \psi_b \rangle \langle \psi_b | q \rangle = |\langle \psi_a | \psi_b \rangle|^2$$

In terms of the Wigner function the above quantity can be written as

$$\text{Tr}(\hat{\rho}_a \hat{\rho}_b) = \int dq dp W_a(q, p) \rho_b(q, p)$$

But since $\rho(q, p) = W(q, p)/2\pi\hbar$ it becomes

$$\text{Tr}(\hat{\rho}_a \hat{\rho}_b) = |\langle \psi_a | \psi_b \rangle|^2 = \frac{1}{2\pi\hbar} \int dq dp W_a(q, p) W_b(q, p)$$

If the two states are orthogonal to each other then the inner product would become 0. That immediately implies that for certain regions of the phase space the Wigner function will be negative.

While (2.6) gives the correspondence between an operator and its phase space function, it can also be used to find the phase space representation of the product of two operators $\hat{A}$ and $\hat{B}$. The Weyl transform of the product $\hat{A}\hat{B}$ is:

$$\hat{A}\hat{B} = \frac{1}{(2\pi\hbar)^2} \int d\sigma d\tau dq dp e^{i[(\hat{q}-q)\sigma + (\hat{p}-p)\tau]} \times [A(q, p) (\exp\left(\frac{i\hbar}{2}(\overleftarrow{\partial}_q \overrightarrow{\partial}_p - \overleftarrow{\partial}_p \overrightarrow{\partial}_q)\right)) B(q, p)] \quad (2.8)$$

Thus the corresponding phase space function is defined using something known as the Moyal[2] $\star$ (star) product, i.e.,

$$\hat{A}\hat{B} \rightarrow A(q, p) \star B(q, p) \quad (2.9)$$

The $\star$ operator is given by

$$\star \equiv e^{\frac{i\hbar}{2}(\overleftarrow{\partial}_q \overrightarrow{\partial}_p - \overleftarrow{\partial}_p \overrightarrow{\partial}_q)} \quad (2.10)$$

The $\star$ operator introduced above seems intimidating but for practical calculations there is a way to make

it easier to use. This is done by introducing something known as the Bopp shift which shifts the variables of either of the two functions by a certain amount.

$$\begin{aligned} A(q, p) \star B(q, p) &= A\left(q + \frac{i\hbar}{2} \frac{\partial}{\partial p}, p - \frac{i\hbar}{2} \frac{\partial}{\partial q}\right) B(q, p) \\ &= A(q, p) B\left(q - \frac{i\hbar}{2} \frac{\partial}{\partial p}, p + \frac{i\hbar}{2} \frac{\partial}{\partial q}\right) \end{aligned} \quad (2.11)$$

It is easy to see that the Bopp shifts hold. It can be done by just Taylor expanding the exponential into its Taylor series. This is a Taylor series in 2 variables and thus expanding the exponential upto the first order we get:

$$e^{\frac{i\hbar}{2} (\overleftarrow{\partial}_q \overrightarrow{\partial}_p - \overleftarrow{\partial}_p \overrightarrow{\partial}_q)} = \mathbb{1} + \frac{\overleftarrow{\partial}}{\partial q} \left(\frac{i\hbar}{2} \frac{\overrightarrow{\partial}}{\partial p} \right) - \frac{\overleftarrow{\partial}}{\partial p} \left(\frac{i\hbar}{2} \frac{\overrightarrow{\partial}}{\partial q} \right)$$

The above expression acts on $A(q, p)$ and it is evident that these are the terms upto first order of the translation operator which is of the form $e^{i\alpha\partial_q/\hbar}$ which translates a function $T(q) \rightarrow T(q + \alpha)$. Following that convention we can see that for $A(q, p)$ each left acting derivative shifts $q \rightarrow q + \frac{i\hbar}{2} \partial_p$ and $p \rightarrow p - \frac{i\hbar}{2} \partial_q$. For the 2nd equality we just let the right acting derivatives operate on $B(q, p)$ after expanding and we get the corresponding shift. As an example we can calculate what the product $\hat{q}\hat{p}$ is using the $\star$ -product and the Bopp shifts

$$\begin{aligned} \hat{q}\hat{p} &= q \star p \\ &= \left(q + \frac{i\hbar}{2} \frac{\partial}{\partial p} \right) p \\ &= qp + \frac{i\hbar}{2} \end{aligned}$$

The general expression for the operator expression for a product of the form $q^n p^m$ is given by the Weyl-ordering of operators [10].

$$q^n p^m = \frac{1}{2^n} \sum_{r=0}^n \binom{n}{r} \hat{q}^{n-r} \hat{p}^m \hat{q}^r \quad (2.12)$$

Thus the product qp would be equal to $(\hat{q}\hat{p} + \hat{p}\hat{q})/2$. The relation obtained using the $\star$ -product can be rewritten as:

$$\hat{q}\hat{p} = qp + \frac{i\hbar}{2} = qp + \frac{[\hat{q}, \hat{p}]}{2} = qp + \frac{\hat{q}\hat{p} - \hat{p}\hat{q}}{2}$$

Thus taking this over to the LHS which is $\hat{q}\hat{p}$ gives the Weyl-ordered operator form of the phase space function qp .

2.1 Time Evolution

The time evolution of the Wigner function provides an excellent insight into its relation to classical phase space. It is given by Moyal's equation [2]

$$i\hbar \frac{\partial W}{\partial t} = \{\{H, W\}\}_{MB} = H \star W - W \star H \quad (2.13)$$

Eq. (2.13) provides a compact expression of the time evolution which looks very similar in structure to Liouville's equation. In fact, the Moyal Bracket is nothing but a deformation of the Poisson bracket parametrized by $\hbar$. It can be understood better by expanding the star product. Doing so leads to the equation [9]

$$\frac{\partial W}{\partial t} = -\frac{p}{m} \frac{\partial W(q, p)}{\partial x} + \sum_{s=0}^{\infty} \frac{1}{(2s+1)!} \left(\frac{\hbar}{2i}\right)^{2s} \left(\frac{\partial^{2s+1}}{\partial x^{2s+1}}\right) V(x) \left(\frac{\partial^{2s+1}}{\partial p^{2s+1}}\right) W(q, p) \quad (2.14)$$

The expression is a fairly massive one. However if we set $s = 1$, the first order correction the time evolution is:

$$\frac{\partial W}{\partial t} = -\frac{p}{m} \frac{\partial W(q, p)}{\partial q} + \frac{\partial V}{\partial q} \frac{\partial W}{\partial p} - \frac{\hbar}{24} \frac{\partial^3 V}{\partial q^3} \frac{\partial^3 W}{\partial p^3}$$

This equation is nothing but a correction to Liouville's equation in terms of $\hbar^2$. Thus, Moyal's equation provides corrections to Liouville's equation in terms of powers of $\hbar^2$. In other words, quantum mechanics introduces volume non-preserving elements, in orders of the reduced Planck's constant to the classical Liouville's equation. While Liouville's theorem implies that the classical volume distribution remains constant over time, the presence of higher order derivatives in Moyal's equation breaks this constraint. The possibility of higher order derivatives implies that the volume of the probability fluid changes over time. If we put $s = 0$ which amounts to also taking $\hbar \rightarrow 0$, i.e., the classical limit, we retrieve Liouville's equation for $W(q, p)$. This is a striking result and provides a glimpse into the connection between classical and quantum mechanics. Interestingly, it is easy to see that for quadratic Hamiltonians, such as the harmonic oscillator, the quantum version and the classical version are the same which is a neat fact and shows that quadratic Hamiltonians are really special since both classical and quantum mechanics essentially treat them the same way in terms of time evolution.

From a mathematical perspective it is important to establish the equivalence of eqs. (2.13) and (2.14). To find the consequence of the $\star$ operator, the Bopp shift can be applied to both H and W . Since the Hamiltonian is a sum of $\hat{q}$ and $\hat{p}$ operators with no operator products its Weyl symbol can be obtained

by just replacing the operators with q and p . Thus the terms in the Moyal equation that have the star product become

$$\begin{aligned} W \star H &= H \left(x - \frac{i\hbar}{2} \frac{\partial}{\partial p}, p + \frac{i\hbar}{2} \frac{\partial}{\partial x} \right) W(q, p) \\ &= \left[\frac{1}{2m} \left(p + \frac{i\hbar}{2} \frac{\partial}{\partial x} \right)^2 + V \left(x - \frac{i\hbar}{2} \frac{\partial}{\partial p} \right) \right] W(q, p) \end{aligned}$$

$$\begin{aligned} H \star W &= H \left(q + \frac{i\hbar}{2} \frac{\partial}{\partial p}, p - \frac{i\hbar}{2} \frac{\partial}{\partial q} \right) W(q, p) \\ &= \left[\frac{1}{2m} \left(p - \frac{i\hbar}{2} \frac{\partial}{\partial q} \right)^2 + V \left(q + \frac{i\hbar}{2} \frac{\partial}{\partial p} \right) \right] W(q, p) \end{aligned}$$

Plugging this into eq. (2.13) gives

$$\frac{\partial W}{\partial t} = \frac{i}{\hbar} \left[\frac{1}{2m} \left\{ \left(p + \frac{i\hbar}{2} \frac{\partial}{\partial q} \right)^2 - \left(p - \frac{i\hbar}{2} \frac{\partial}{\partial q} \right)^2 \right\} + \left\{ V \left(q - \frac{i\hbar}{2} \frac{\partial}{\partial p} \right) - V \left(q + \frac{i\hbar}{2} \frac{\partial}{\partial p} \right) \right\} \right] W(q, p)$$

The momentum part is just the same formula as $a^2 - b^2$ where $a = p + \frac{i\hbar}{2} \frac{\partial}{\partial q}$ and $b = p - \frac{i\hbar}{2} \frac{\partial}{\partial q}$. Then it eventually becomes

$$\frac{i}{\hbar} \left[\frac{1}{2m} \left(p + \frac{i\hbar}{2} \frac{\partial}{\partial q} \right)^2 - \left(p - \frac{i\hbar}{2} \frac{\partial}{\partial q} \right)^2 \right] = -\frac{p}{m} \frac{\partial}{\partial q}$$

The potential term can be Taylor expanded and written out in the same way as in eq. (2.14). Without performing the involved algebra it can be seen that the Taylor expansion of this potential term will give the same as it did in eq. (2.14). In the derivation of that equation if we just replace y with $-i\hbar \partial_p$ then we can see:

$$\begin{aligned} V \left(q - \frac{i\hbar}{2} \frac{\partial}{\partial p} \right) - V \left(q + \frac{i\hbar}{2} \frac{\partial}{\partial p} \right) &= \sum_{n=0}^{\infty} \frac{1}{n!} V^{(2s+1)}(q) \left(\left(-\frac{i\hbar}{2} \frac{\partial}{\partial p} \right)^n - \left(\frac{i\hbar}{2} \frac{\partial}{\partial p} \right)^n \right) \\ &= \sum_{s=0}^{\infty} \frac{1}{(2s+1)!} V^{(2s+1)}(x) \frac{1}{2^{2s}} \left(\frac{\hbar}{i} \right)^{2s+1} \frac{\partial^{2s+1}}{\partial p^{2s+1}} \end{aligned}$$

Thus multiplying the above relation with the factor $i/\hbar$ and operating on $W(q, p)$ gives back eq. (2.14) thus establishing the equivalence of Moyal's equation (eq. (2.13)) and eq. (2.14).

Chapter 3

Wigner function for discrete systems

In chapter 2 the definition of the Wigner function was provided for systems that have an infinite dimensional Hilbert space. However, systems with finite Hilbert spaces are also common in quantum mechanics. Wootters [5] provided a consistent formulation of the Wigner function in terms of finite sized Hilbert spaces. Under Wootters' formulation the phase space is converted from a continuous Cartesian grid to a lattice of points. The number of points in each direction is equal to the Hilbert space dimension, N . The coordinates are equal to the eigenvalues of the observable that is associated with each axis. To make matters simple, the coordinates are labelled from 0 to $N - 1$.

The mathematical analysis of the finite state Wigner function becomes simple for N a prime number. Prime N -s are analogous to one dimensional systems in the continuous case, i.e., these cannot be subdivided into composite subsystems. Thus, composite systems, i.e., systems that have N not prime are treated similar to how higher dimensional systems are treated in the continuous case. The phase space for system with N not prime are taken as Cartesian products of the smaller systems with N as prime. In other words systems of N not equal to a prime number are considered as composite systems of the prime factors of N .

3.1 Phase Point operators

The projection property that was mentioned in (2.3) and (2.4) is a special case of a more general property. The general projection property states that the integral over phase space of the Wigner function bounded by two parallel lines $c_1 \leq aq + bp \leq c_2$ is equal to the probability that the observable $a\hat{x} + b\hat{p}$ has an

expectation value between c_1 and c_2 . Mathematically this property can be written as

$$\int \int_{c_1 \leq aq + bp \leq c_2} W(q, p) dq dp = \text{Tr}(\hat{\rho}(a\hat{x} + b\hat{p})) \quad (3.1)$$

This value lies in the range $[c_1, c_2]$

The connection between the Wigner function and the density matrix can be states in terms of a phase point operator as well, instead of a Fourier transform. In terms of the phase point operators, $\hat{A}(q, p)$, the density matrix can be expressed:

$$\hat{\rho} = \int W(q, p) \hat{A}(q, p) dq dp \quad (3.2)$$

Eq. (3.2) shows that the density matrix can be expanded in a parametrized basis which is spanned by the phase point operators. The coefficients of this expansion is the Wigner function. The phase point operator has a few properties which can be summarized as follows.

Property A1: For each point in phase space (q, p) , $\text{Tr}(\hat{A}(q, p)) = 1$.

Property A2: For pairs of points (q_1, p_1) and (q_2, p_2)

$$\text{Tr}(\hat{A}(q_1, p_1) \hat{A}(q_2, p_2)) = 2\pi\hbar \delta(q_1 - q_2) \delta(p_1 - p_2)$$

Property A3: Consider a family of parallel lines given by $c_1 \leq aq + bp \leq c_2$ then the projection operator $\hat{P}$ is constructed as

$$\hat{P} = \frac{1}{2\pi\hbar} \int \int_{c_1 \leq aq + bp \leq c_2} \hat{A}(q, p) dq dp \quad (3.3)$$

The operator $\hat{P}$ projects onto the subspace spanned by the eigenstates of the observable $a\hat{q} + b\hat{p}$ with eigenvalues lying between c_1 and c_2 .

The Wigner function can be obtained in terms of $\hat{A}(q, p)$ by using Property A2 in eq. (3.2).

$$W(q, p) = \frac{1}{2\pi\hbar} \text{Tr}(\hat{\rho} \hat{A}(q, p)) \quad (3.4)$$

3.2 Phase space for finite dimensional systems

The discrete phase space is a square grid of $N \times N$ dimensions where N is the number of eigenstates of each observable associated with the axes of the space. The simplest systems to consider are those where N is a prime number. The axes in each direction take values going from 0 to $N - 1$. The integers ranging

from 0 to $N - 1$ form a finite field, $\mathbb{Z}_N$ and the phase space is a 2 dimensional vector space, $P = \mathbb{Z}_N^2$ defined over this field. It is the finite, discrete analogue of $\mathbb{R}^2$. The reason why N has to be prime to represent a single system is to ensure that $\mathbb{Z}_N^2$ forms a field and the geometry in this system remains the same as in a continuous grid. All mathematical operations are performed $\text{mod } N$. If N is non-prime it is possible that the intersections between two “lines” in this phase space will have more than 1 unique solution. That would lead to an ambiguous geometry which is not desirable. This is just one example of the problems that would arise due to a non-prime N .

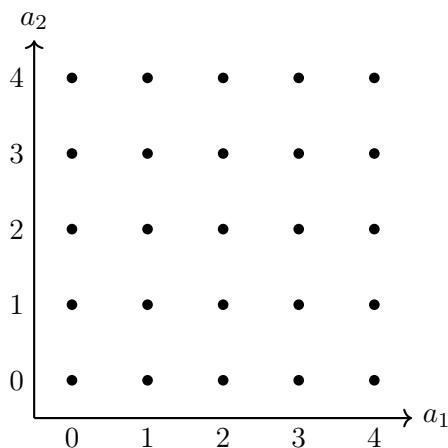


Figure 3.1: Phase space for system with 5 eigenstates

In the same way that lines are described in $\mathbb{R}^2$, lines and parallel lines are described in $\mathbb{Z}_N^2$. A line is defined as $ma_1 + na_2 = p$ where m , n and p belong to $\mathbb{Z}_N$ and all additions and multiplication operations occur modulo N . There are a total of $N(N + 1)$ lines in the phase space. These lines can be thought of as $N + 1$ sets of N parallel lines. In the context of the discrete phase space a complete set of N parallel lines is known as a “foliation”.

The notion of area in $\mathbb{Z}_N^2$ is a little different from $\mathbb{R}^2$. The notion of area in $\mathbb{R}^2$ is intimately tied to the area of a triangle. Any simple polygon in $\mathbb{R}^2$ can be decomposed into multiple triangles [11]. Thus the area of any simple shape can be calculated by summing the areas of its constituent triangles. The area in $\mathbb{Z}_N^2$ cannot be simply defined using lengths since distance between points do not really mean anything in this space. Thus the area of triangle is defined with reference to a standard triangle of unit area. A triangle of unit area is one which has is defined by the points $\{(0,0), (1,0), (0,1)\}$. This triangle is defined to have unit area. All other possible triangles $\{\alpha, \beta, \gamma\}$ can be defined as affine transformations of the unit triangle as $a'_i = \sum_j M_{ij}a_j + C_i$ where a'_i is the new vertex, a_j is the original triangle, M_{ij} is the transformation matrix and C_i is a translation vector. The area of the new triangle is equal to the determinant of the

matrix M .

The labelling of the axes in this space is similar to how it is done in the continuous case. In the continuous case the axes are said to be labelled by the observables $\hat{q}$ and $\hat{p}$. What that really means is that the vertical lines are associated with those observables but are labelled by the eigenvalues of each observable. The scenario is the same for the discrete phase space. While there is a great deal of freedom in choosing observables there is one constraint. They have to be mutually unbiased. Two orthonormal bases are called mutually unbiased if the inner product of the basis vectors gives a uniform probability measure that does not depend on the specific basis vector chosen. In mathematical terms if A is an observable with the eigenstates $|a_i\rangle_{i=0}^{N-1}$ and B is another observable with the eigenstates $|b_j\rangle_{j=0}^{N-1}$ then these two are mutually unbiased iff:

$$|\langle a_i | b_j \rangle|^2 = \frac{1}{N}$$

It is important at this point to consider systems for which N is not a prime number. In that case the phase space of the system is represented as the Cartesian product of systems which are prime factors of N . In other words the composite system is considered to be constituted of individual subsystems which have dimensions equal to the prime factors of N . For example, a spin-1/2 and spin-1 particle composite system would have $N = 6$. Its phase space would be a 6×6 square grid of points. The phase space can be thought of as $2 \times 2 \times 3 \times 3$ as well. Thus specifying a point in the composite phase space is equivalent to simultaneously specifying a point in the constituent subsystem phase spaces.

3.3 Wigner function of finite dimensional systems

The properties and the definition of the phase space operator remains largely the same in the discrete case as it was in the continuous case. Slight changes occur in the mathematical form as a result of going from the continuous to the discrete case. For each point in the N^2 dimensional phase space there are N^2 phase point operator, $\hat{A}_\alpha$ associated with each point α . Each of these operators is an $N \times N$ matrix. The properties of the discrete phase point operator are:

Property A1': For each point, α , $\text{Tr}(\hat{A}_\alpha) = 1$

Property A2': For two different points, α and β , $\text{Tr}(\hat{A}_\alpha \hat{A}_\beta) = N\delta_{\alpha\beta}$

Property A3': Consider a set of N parallel lines. For each line λ a projection operator can be constructed as

$$\hat{P}_\lambda = \frac{1}{N} \sum_{\alpha \in \lambda} \hat{A}_\alpha \quad (3.5)$$

Phase point operators that have all three of the properties mentioned above have simple constructions. These are different for $N = 2$ and for prime $N > 2$. For $N = 2$ it is

$$\hat{A}_\alpha = \frac{1}{2} [(-1)^{a_1} \hat{\sigma}_z + (-1)^{a_2} \hat{\sigma}_x + (-1)^{a_1+a_2} \hat{\sigma}_y + \hat{\mathbb{1}}] \quad (3.6)$$

And for prime $N > 2$ it is

$$(A_\alpha)_{kl} = \delta_{2a_1, k+l} e^{(2i\pi/N)a_2(k-l)} \quad (3.7)$$

The variables a_1 and a_2 refer to the horizontal and vertical coordinates of the point α . In eq. (3.7) the indices k and l refer to the matrix indices of the operator $\hat{A}_\alpha$. For a composite value of N the phase point operators can be defined by decomposing N into its prime factors, $N_1, \dots, N_f$. Given that for each N_i there is a separate phase space then a point on the composite phase space can be thought of as a sort of collection of points on the separate sub-phase spaces, $\alpha = (\alpha_1, \dots, \alpha_f)$. The phase point operator for the composite system is a tensor product of the phase point operators of all subsystems

$$\hat{A}_\alpha = \hat{A}_1 \otimes \dots \otimes \hat{A}_f \quad (3.8)$$

The density matrix, $\hat{\rho}$ is the representation of a system with N eigenstates. Property (A1') and (A2') imply that the set of phase point operators, $\hat{A}_\alpha$, form an orthonormal basis for representing the density matrix of the system. In other words the density matrix can be written as a linear combination of the set of phase point operators. The Wigner function is defined as the coefficients in the linear combination.

$$\hat{\rho} = \sum_{\alpha} W_{\alpha} \hat{A}_{\alpha} \quad (3.9)$$

Using Property (A2') eq. (3.9) can be inverted to obtain the relation for the Wigner function as

$$W_{\alpha} = \frac{1}{N} \text{Tr}(\hat{\rho} \hat{A}_{\alpha}) \quad (3.10)$$

The properties of the Wigner function are similar to that in the continuous case and it is instructive to list all the main properties here.

Property W1':

$$\text{Tr}(W_{\alpha}) = 1$$

Property W2': If W and W' are two Wigner functions corresponding to density matrices, $\hat{\rho}$ and $\hat{\rho}'$ then,

$$N \sum_{\alpha} W_{\alpha} W'_{\alpha} = \text{Tr}(\hat{\rho} \hat{\rho}') \quad (3.11)$$

This is similar to the continuous case. It implies that if the two states are orthogonal then the RHS is equal to 0. Just like in the continuous case this implies that for some points α the Wigner function has negative values.

Property W3': Consider the phase space Φ to be foliated into N parallel slices and take any line λ from that set then the sum $p_{\lambda} = \sum_{\alpha \in \lambda} W_{\alpha}$ represents the probability of the outcome of a specific measurement associated with the particular set of parallel slices that have been chosen.

In this case a measurement refers to the entire set of eigenstates associated that are represented by foliations. Thus what property (W3') is saying is that the sum of the values of of the Wigner function over a chosen slice in the phase space is equal to the probability of the system being in the particular eigenstate given that it was prepared in the specific state that was used to construct the Wigner function. In more mathematical terms, remembering that an eigenstate has the projector defined by Property (A1') and eq. (3.5) then we have

$$\text{Tr}(\hat{\rho} \hat{P}_{\lambda}) = \sum_{\alpha \in \lambda} W_{\alpha} \quad (3.12)$$

The operator $\hat{P}_{\lambda}$, in essence, filters out the the eigenstates from the density matrix that do not constitute the projector itself. A trace over that quantity is a sum of diagonals and thus represents the probability that the system is in that particular state given it was prepared in another state earlier.

The properties become clear when considering a real system. The simplest system that can be considered is a spin-1/2 system which will have only a 2×2 phase space. The axes will be associated with $\hat{\sigma}_z$ in the a_1 direction and $\hat{\sigma}_x$ in the a_2 direction. Using eq. (3.6) the phase point operators for each point of the phase space is given by

$$\begin{aligned} A_{01} &= \frac{1}{2} \begin{pmatrix} 2 & -1+i \\ -1-i & 0 \end{pmatrix} & A_{11} &= \frac{1}{2} \begin{pmatrix} 0 & -1-i \\ -1+i & 2 \end{pmatrix} \\ A_{00} &= \frac{1}{2} \begin{pmatrix} 2 & 1-i \\ 1+i & 0 \end{pmatrix} & A_{10} &= \frac{1}{2} \begin{pmatrix} 0 & 1+i \\ 1-i & 2 \end{pmatrix} \end{aligned} \quad (3.13)$$

Using eq. (3.13) we can construct the Wigner function for different states using eq. (3.10). As an

illustration of the Properties (W1') - (W3') we compute two separate Wigner functions for two states. Consider the +1 eigenstate of both $\hat{\sigma}_z$ and $\hat{\sigma}_x$. They are given by $|0\rangle = \begin{pmatrix} 1 & 0 \end{pmatrix}^T$ and $|+\rangle = 1/\sqrt{2} \begin{pmatrix} 1 & 1 \end{pmatrix}^T$ (the values in kets are just for representation). Thus the density matrices for each are given by

$$\hat{\rho}_{|0\rangle} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\hat{\rho}_{|+\rangle} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Using these the Wigner functions for each are:

$$W_{|0\rangle} :$$

$\frac{1}{2}$	0
$\frac{1}{2}$	0

$$W_{|+\rangle} :$$

0	0
$\frac{1}{2}$	$\frac{1}{2}$

Property (W1') is very evident in both of the above since the sum over all points is equal to 1 as expected. As for Property (W2') again it is very evident that $\text{Tr}(\hat{\rho}_{|0\rangle}\hat{\rho}_{|+\rangle}) = 1/2$ and $N \sum_{\alpha} W_{|0\rangle} W_{|+\rangle} = 1/2$. For property (W3') we can consider only $W_{|0\rangle}$ and the same results will hold for $W_{|+\rangle}$.

There are 3 different sets of 2 parallel lines. If we pick the two horizontal lines, $a_2 = 0$ and 1, and sum over each we get 1/2. The two sets of horizontal lines correspond to the eigenstates of the $\hat{\sigma}_x$ which can be written as a linear combination of the eigenstates of $\hat{\sigma}_z$ as $(|0\rangle \pm |1\rangle)/\sqrt{2}$ where $|1\rangle = \begin{pmatrix} 0 & 1 \end{pmatrix}^T$. Thus it is evident that the probability of the system being in either eigenstate of $\hat{\sigma}_x$ is 1/2. The same idea applies if we pick either diagonal which represent both the eigenstates of $\hat{\sigma}_y$. The distinction lies when either of $a_1 = 0$ or $a_1 = 1$ lines are picked. The sum over $a_1 = 0$ gives 1 while the sum over $a_1 = 1$ is 0. This is to be expected since the system has been prepared in the $|0\rangle$ eigenstate of $\hat{\sigma}_z$ so the probability that a measurement gives $|0\rangle$ as the outcome is 1 while for $|1\rangle$ is 0. This is Property (W3'). The same will hold for the case of $W_{|+\rangle}$.

The observables that have the eigenvalues a_1 and a_2 can be constructed from the projectors corresponding to the eigenstate that a particular parallel line in phase space corresponds to. For the $N = 2$ case the observables can be written as

$$\begin{aligned}\hat{O}_1 &= \frac{1}{N} \sum_{\alpha} a_1 \hat{A}_{\alpha} \\ \hat{O}_2 &= \frac{1}{N} \sum_{\alpha} a_2 \hat{A}_{\alpha}\end{aligned}\tag{3.14}$$

For the $N > 2$ prime case the matrix elements of the observables can be written as

$$\begin{aligned}(O_1)_{kl} &= k\delta_{kl} \\ (O_2)_{kl} &= \frac{1}{N} \sum_{a_2} a_2 e^{(2i\pi/N)a_2(k-l)}\end{aligned}\tag{3.15}$$

In general any operator $\hat{O}$ can be expressed in the discrete phase space representation in a way similar to the density matrix. One of the definitions of the discrete Wigner transform and the corresponding Weyl transform is

$$\hat{O} = \frac{1}{N} \sum_{\alpha} O_{\alpha} \hat{A}_{\alpha} \quad O_{\alpha} = \text{Tr}(\hat{O} \hat{A}_{\alpha})\tag{3.16}$$

It is evident that in defining the above quantities the factor of $1/N$ is in equation for $\hat{O}$ and not for its phase space equivalent. This was opposite in the case of the Wigner function. We can also write it either way but the important thing to note is that the normalization factor that we get by multiplying both factors outside each transform should be equal to $1/N$. This is a consequence of the orthogonality condition of the phase point operators as in Property (A2').

3.4 Laws of quantum mechanics in terms of Wigner function

Some very basic postulates of quantum mechanics can be written for finite dimensional systems in terms of the density matrix. These postulates can then be recast in the language of the Wigner function using the defined relation between the density matrix and Wigner function for finite dimensional systems.

The basic postulates in terms of the density matrix are:

- (a) *Representation of states*: Every state is represented by an $N \times N$ Hermitian matrix $\hat{\rho}$ with $\text{Tr}(\hat{\rho}) = 1$.

For pure states, i.e., density matrices constructed from eigenstates of the Hamiltonian, $\hat{\rho} = \hat{\rho}^2$. Mixed states are states created using weighted averages of pure states.

- (b) *Expectation values*: Observables are represented by operators which are $N \times N$ Hermitian matrices.

The expectation value of an observable defined by the operator $\hat{O}$ in a particular state, $\hat{\rho}$ is given by $\langle O \rangle = \text{Tr}(\hat{\rho}\hat{O})$.

- (c) *Time evolution*: The time evolution of the states is related to the Hamiltonian, $\hat{H}$ of the system through the relation

$$\frac{d\hat{\rho}}{dt} = -\frac{i}{\hbar}[\hat{H}, \hat{\rho}] \quad (3.17)$$

To reconstruct the above laws in terms of the discrete Wigner function a new quantity constructed from phase point operators of 3 different points in phase space can be defined:

$$\Gamma_{\alpha\beta\gamma} = \frac{1}{N} \text{Tr}(\hat{A}_\alpha \hat{A}_\beta \hat{A}_\gamma) \quad (3.18)$$

The laws can now be written in terms of the Wigner function as:

- (a') A pure state Wigner function, W_α , satisfies the relation

$$W_\alpha = \sum_\beta \sum_\gamma \Gamma_{\alpha\beta\gamma} W_\beta W_\gamma \quad (3.19)$$

Mixed states are represented as weighted sums of the pure states.

- (b') Given an observable which is defined as $\hat{O} = (1/N) \sum_\alpha O_\alpha \hat{A}_\alpha$ the expectation value of the observable in a given state is

$$\langle O \rangle = \sum_\alpha W_\alpha O_\alpha \quad (3.20)$$

- (c') The evolution of a state, W_α can be written in terms of the Hamiltonian defined as $\hat{H} = \sum_\alpha H_\alpha \hat{A}_\alpha$

$$\frac{dW_\alpha}{dt} = -\frac{i}{\hbar} \sum_\beta \sum_\gamma \Gamma_{\alpha\beta\gamma} (H_\beta W_\gamma - W_\beta H_\gamma) \quad (3.21)$$

This is the discrete analogue of the Moyal equation.

The underlying structure of the formulation is expressed in the quantity, $\Gamma_{\alpha\beta\gamma}$. In the case of prime $N > 2$ the expression for this quantity is

$$\Gamma_{\alpha\beta\gamma} = \frac{1}{N} \exp\left\{\left(\frac{4i\pi}{N} \mathcal{A}_{\alpha\beta\gamma}\right)\right\} \quad (3.22)$$

Where $\mathcal{A}_{\alpha\beta\gamma}$ is the area of the triangle characterized by the three point, $\{\alpha, \beta, \gamma\}$. In the case of $N = 2$

the relation is

$$\Gamma_{\alpha\beta\gamma} = \delta_{\alpha\beta} + \delta_{\beta\gamma} + \delta_{\gamma\alpha} - \frac{1}{2} - i \sum_{\delta} \varepsilon_{\alpha\beta\gamma\delta} \quad (3.23)$$

Here the term $\varepsilon_{\alpha\beta\gamma\delta}$ is antisymmetric in all of its indices, i.e., its sign flips if any pair of indices are swapped. It is equal to 0 if any of the indices are the same and finally $\varepsilon_{\alpha\beta\gamma\delta} = 1$ if the ordering of the indices is $\{\alpha, \beta, \gamma, \delta\} = \{(0,0), (1,0), (1,1), (0,1)\}$. This order follows the natural order of square grid in the anti-clockwise direction. Flipping the order of movement along this square flips the sign of $\varepsilon_{\alpha\beta\gamma\delta}$. For composite systems the composite Γ is equivalent to the product of the Γ for each individual subsystem. The time evolution law given by eq. (3.21) can be expressed in another way just like the transformation of $\hat{\rho}$ are encoded by unitary matrices. In mathematical terms transformations that evolves the density matrix in time is the transformation $\hat{\rho} \rightarrow \hat{U}^\dagger \hat{\rho} \hat{U}$ where $\hat{U}$ is a unitary operator. The analogous statement in the discrete Wigner function case is $W_\alpha \rightarrow \sum_\beta Z_{\alpha\beta} W_\beta$. The conditions on $Z_{\alpha\beta}$ are:

1. Z is real
2. $\sum_\alpha Z_{\alpha\beta} = 1$
3. $\sum_{\rho, \sigma, \tau} Z_{\alpha\rho} Z_{\beta\sigma} Z_{\gamma\tau} \Gamma_{\rho\sigma\tau} = \Gamma_{\alpha\beta\gamma}$

These 3 statements say that Z must preserve the normalization and reality of W and also keep Γ invariant.

Chapter 4

The Quantum Kicked Rotor on a Torus

The kicked rotor is one of the most well studied classical maps which shows a clear transition to chaotic behaviour on increasing the values of parameters of the system. The Chirikov map, or the standard map, as the classical system is called is defined on a cylinder and is given by the Hamiltonian $H = p^2/2m + K \cos(q) \sum_n \delta(t - nT)$ which when applying Hamilton's equations gives the following iterative map

$$p_{n+1} = p_n + K \sin(q_n) \quad q_{n+1} = q_n + p_{n+1} \quad (4.1)$$

This system was first proposed and studied in detail by Chirikov [12]. It is a very simple system that shows chaotic behaviour in certain value of K . The phase space is a cylinder since the position is constrained in $[0, 2\pi]$. It is obvious that for $K \approx 0$ the system is integrable. The introduction of non-zero K introduces a perturbation to the system which leads to the breaking of periodic and quasi-periodic tori. In accordance with the KAM theorem of classical mechanics, which states that quasiperiodic tori remain unbroken upto a certain value of the perturbation introduced to an integrable system, upto a certain value of K , the phase space contains periodic islands. However for values of $K > 5$, the phase space becomes entirely mixed and no semblance of periodic tori remain.

Quantization of the system and subsequent study of the evolution operator provides a means of studying chaos in quantum systems. The Hamiltonian of this system is defined as [13]¹

$$\hat{H} = \frac{\hat{p}^2}{2m} - \frac{\kappa}{4\pi^2} \cos(2\pi\hat{q}) \sum_n \delta(t - nT) \quad (4.2)$$

¹The constants of the system like m , the mass, and the time period, T have been set to 1 for the rest of the section and for all computations as well. In eq. (4.8) the constants are shown but beyond that they are set to 1 everywhere.

Where κ is the kicking strength, T is the time period. Floquet theory can be used to understand the time evolution of this system given it is periodically driven. As a general note every operation that is performed for this system is performed modulo N to preserve the periodic structure of the torus.

4.1 Torus as a phase space

The phase space in this system is defined by $(q, p) \in [0, 1) \times [0, 1) \equiv \mathbb{T}^2$ where q and p are eigenvalues of the position and momentum operators. Periodic boundary conditions imply that for position of momentum eigenstates, with eigenvalues q and p respectively, labelled as $|q\rangle$ and $|p\rangle$ respectively, $|q+1\rangle = |q\rangle$ and $|p+1\rangle = |p\rangle$. This further implies that both position and momentum eigenvalues are quantized as $q = nh$ and $p = mh$ with $0 \leq m, n \leq 1/h$. Since it is known that n and m are integers we can say that the value of each cannot exceed a certain integer N . In other words the eigenvalues take the form $q = n/N$ and $p = m/N$. It is evident from this replacement that on increasing N the number of eigenvalues themselves increase. In other words each "cell" in $\mathbb{T}^2$ becomes smaller and the "resolution" of the space increases. This integer N effectively is the Hilbert space dimension of the system. Again it is very evident that the limit $N \rightarrow \infty$ is the classical limit. Thus the quantized form of the kicked rotor on a torus is a brilliant system to probe both chaos and the quantum-classical correspondence.

Since the phase space of this system is quantized and finite dimensional, the usual Heisenberg matrix algebra does not work in this space [14]. In other words the canonical commutation relation, $[\hat{q}, \hat{p}] = i\hbar$ does not hold meaning $\mathbb{T}^2$. However, since the eigenvalues that make up $\mathbb{T}$ are quantized a unitary operator basis can be created that generates all possible transformations in the space [14]. In the case of the kicked rotor on a torus such a basis can be created by defining translation operators for both $\hat{q}$, $\hat{T}_q$, and $\hat{p}$, $\hat{T}_p$. The properties of these operators are as follows [15].

$$\hat{T}_p |p\rangle = |p+1\rangle, \hat{T}_q |q\rangle = |q+1\rangle \quad (4.3a)$$

$$\hat{T}_p^N |p\rangle = \exp(-2i\pi\beta) |p\rangle \quad (4.3b)$$

$$\begin{aligned} \hat{T}_q^N |q\rangle &= \exp(2i\pi\alpha) |q\rangle \\ \hat{T}_p |q\rangle &= \exp\left(\frac{2i\pi}{N}(n+\alpha)\right) |q\rangle \\ \hat{T}_q |p\rangle &= \exp\left(-\frac{2i\pi}{N}(m+\beta)\right) |p\rangle \end{aligned} \quad (4.3c)$$

The discrete analogue of the canonical commutation relation is expressed through the noncommutativity

of the two translation operators. Schwinger (1960) first described this relation in terms of the clock and shift operators that he developed as examples of the unitary operator basis. Since the translation operators described above form a basis described by Schwinger they satisfy the relation [14]

$$\hat{T}_q \hat{T}_p = \exp\left(\frac{2i\pi}{N}\right) \hat{T}_p \hat{T}_q \quad (4.4)$$

The parameters α and β control parity and time reversal symmetry respectively [13]. They are introduced to modify the boundary conditions of the system. With these introduced the eigenvalues becomes $q = (n + \alpha)/N$ and $p = (m + \beta)/N$. The range of α and β are both $[0, 1)$. Special cases of boundary conditions are when either α or β are equal to 0 or $1/2$. 0 corresponds to symmetric boundary conditions while $1/2$ corresponds to anti-symmetric boundary conditions. All other values give mixed boundary conditions.

4.2 Floquet theory

The Hamiltonian that describes the kicked rotor is periodically driven. Studying the dynamics of such systems requires using Floquet theory. Given a Hamiltonian which can be written in the general form

$$\hat{H} = \hat{H}_0 + \hat{V} \sum_{n=-\infty}^{\infty} \delta(t - nT) \quad (4.5)$$

Floquet theory allows decomposing the unitary time evolution operator, $\hat{U}$ into a “kick” and a “free” part. The derivation is provided in the *Appendix*.

Thus the Hamiltonian in eq. (4.5) can be factorized and written as [15]

$$\hat{U} = \hat{U}_{\text{kick}} \hat{U}_{\text{free}} = \exp\left(-i\hat{V}T/\hbar\right) \exp\left(-i\hat{H}_0 T/\hbar\right) \quad (4.6)$$

This evolves any state over one time period, T . The evolution over multiple time periods can be derived by exploiting the factorization property [16] $\hat{U}(t_2, t_1) = \hat{U}(t_2, t') \hat{U}(t', t_1)$ (which means that a state has evolved from time t_1 to t_2 . This implies that $\hat{U}(t_0 + 2T, t_0) = \hat{U}(t_0 + 2T, t_0 + T) \hat{U}(t_0 + T, t_0) = [\hat{U}(t_0 + T, t_0)]^2$. This implies that evolution of a system under the effect of time periodic Hamiltonian over multiple time periods can be generalized to

$$\hat{U}(t_0 + nT, t_0) = [\hat{U}(t_0 + T, t_0)]^n \quad (4.7)$$

Thus in the case of the kicked rotor on a torus the evolution the system with an initial state $|\psi_0\rangle$ to

another state $|\psi_n\rangle$ at a time nT is given by

$$|\psi_n\rangle = (\hat{U})^n |\psi_0\rangle \quad (4.8)$$

4.3 Wigner function of the Floquet operator

The unitary time evolution operator for the kicked rotor on a torus can be represented by the following expression.

$$\hat{U} = \exp(i\kappa \cos(2\pi\hat{q})/4\pi^2\hbar) \exp(-i\hat{p}^2/2\hbar) \quad (4.9)$$

In the position basis the matrix elements, i.e., $\langle n|\hat{U}|n'\rangle$ can be expressed as follows [13][15].

$$U_{nn'} = \frac{1}{N} e^{\frac{iN\kappa}{2\pi} \cos\left(2i\pi \frac{(n+\alpha)}{N}\right)} \sum_{m=0}^{N-1} e^{-\frac{i\pi}{N}(m+\beta)^2} e^{\frac{2i\pi}{N}(m+\beta)(n-n')} \quad (4.10)$$

To obtain this expression a modified form of $\langle n|m\rangle$ is introduced due to the presence of α and β .

$$\langle n|m\rangle = \frac{1}{\sqrt{N}} \exp\left(\frac{2i\pi}{N}(n+\alpha)(m+\beta)\right) \quad (4.11)$$

The expression is quite involved. However, some properties can be inferred for particular parameter values. For values of $\kappa \approx 0$ which corresponds to the classically integrable system, the Floquet matrix reduces to the sum only. Furthermore for $\beta = 0$ this matrix becomes circulant, i.e., the columns are cyclic permutations of each other. A non-zero κ introduces a phase to each element and breaks the circulant structure of $\hat{U}$.

The discrete Wigner function set up by Wootters comes particularly handy in analyzing the characteristics of the quantum map. The Wigner function of the eigenstate of the Floquet operator can be created by diagonalizing it and then using the density matrix built from a particular eigenstate in eq. (3.10). The structures in the Wigner function depend very heavily on the values of the parameters, α , β , N and κ .

Fig. 4.1 shows eigenstate for two values of the kicking parameter as examples. It is very evident that the structure is periodic for $\kappa = 0.01$ which is a near integrable system while the phase space is filled up for $\kappa = 10$. For this value of κ the system occupies the whole phase space which means the probability of the system being in any of the eigenstates of the operators that make up the axes is almost equal. The chaotic properties of this map can also be studied through the level spacing statistics.

A robust and often used diagnostic for signatures of chaos in quantum systems are the statistics of the

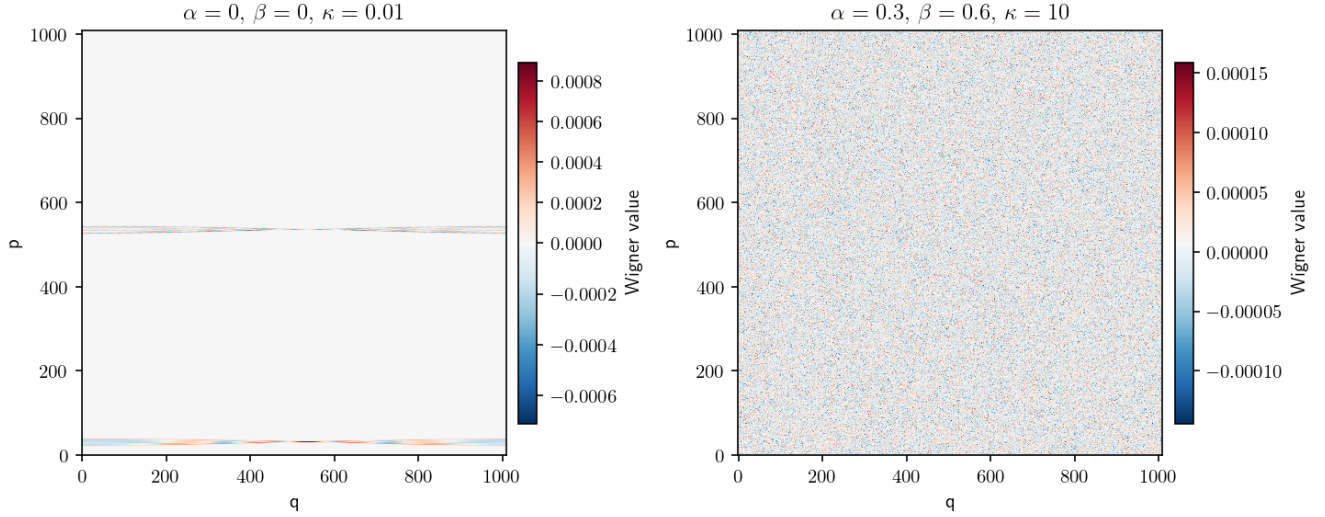


Figure 4.1: Wigner function for the eigenstate corresponding to $\theta = 0.989 - 0.15i$ of the Floquet operator. Construced with $N = 1009$.

spacing of energy levels. The evolution of the kicked rotor is defined by the Floquet operator in eq. (4.10). This operator obeys the eigenvalue relation

$$\hat{U} |\phi_j\rangle = e^{-i\theta_j} |\phi_j\rangle$$

Where ω_j are the quasienergies and are defined modulo N . Level spacing statistics require studying the distribution of unfolded spectrum of the Floquet operator. Unfolding refers to dividing the entire spectrum by the mean so that average spacing is equal to 1. The relation between quantum chaos and the level spacing statistics comes from elements of Random Matrix Theory (RMT). RMT was first used by Wigner to explain the energy level spacings in nuclear spectra. The fundamental idea behind it was to model the Hamiltonian of complex many-body systems as random matrices and study the eigenvalue properties of

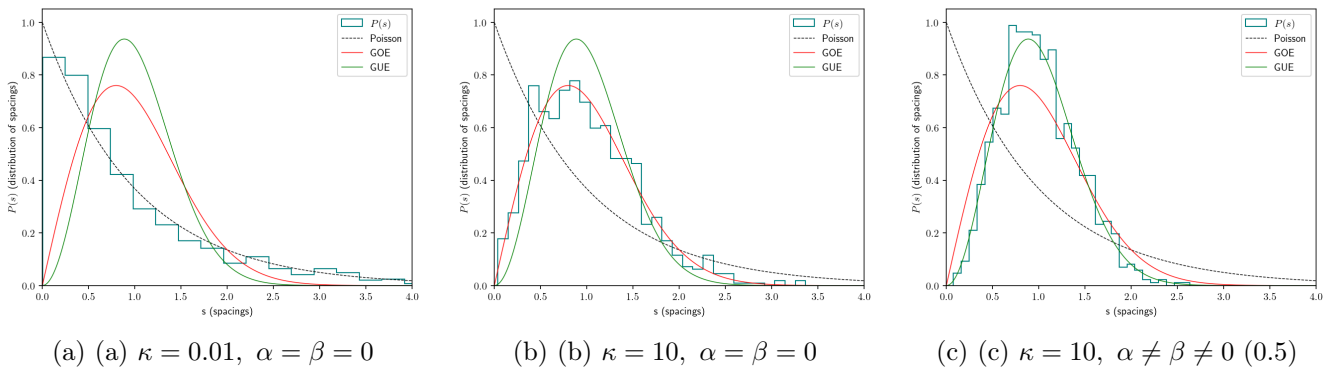


Figure 4.2: Level spacing statistics for $N = 1009$.

such matrices. Within RMT the most commonly encountered classes of random matrices in the study of quantum chaos are Gaussian Orthogonal Ensembles (GOE) and Gaussian Unitary Ensembles (GUE). The kind of ensemble is defined by which transformation leaves the random matrices unchanged. Furthermore Gaussian implies that the entries of the matrix are drawn from a normal distribution [15].

The level spacing statistics for the unfolded spectrum, s for different ensembles are given by

$$P_{\text{Poisson}}(s) = e^{-s} \quad (4.12a)$$

$$P_{\text{GOE}}(s) = \frac{\pi}{2} s e^{-\frac{\pi}{4} s^2} \quad (4.12b)$$

$$P_{\text{GUE}}(s) = \frac{32}{\pi^2} s^2 e^{-\frac{4}{\pi} s^2} \quad (4.12c)$$

The GOE ensemble is seen for systems that exhibit chaos but are time reversal symmetric[3] while GUE is seen for those that are not time reversal symmetric [4].

These relations can be very easily confirmed by plotting the level spacing statistics for different parameter values of the kicked rotor. Fig. 4.2 show 3 different sets of parameter values. The distributions of spacings in the unfolded spectrum follow the expected distributions according to features of different ensembles from RMT. For any value of $\beta \neq \{0, 1/2\}$ time reversal symmetry is broken which means that the system follows the GUE level spacings.

Chapter 5

Wigner function negativity

The Wigner function, as we have seen, is not a strict probability distribution since it allows for negative values. The negativity of the Wigner function can be used as a measure of how non-classical a quantum state is. Nonclassicality is a useful metric in the study of pure quantum states since a class of states, known as coherent states exist, which realize the lower bound of the Heisenberg uncertainty principle. As a result these states are considered as analogues of points in classical phase space. There are two ways in which the negativity of the Wigner function provides a quantitative measure of the non-classicality of a pure quantum state. The first is the fact that the Wigner function of coherent states are positive everywhere. Thus, the measure of how nonclassical a state is characterized by the negative volume of the Wigner function [17]. Another link comes from the Gottesmann-Knill theorem [18] which says that any quantum circuits which initialize to a positive Wigner function state or quantum operations with a positive Wigner function can be efficiently simulated on a classical computer. In other words, positive Wigner functions are an indication of states that are the most “classical”.

The negativity of the Wigner function has multiple different definitions. In this work the definition used was:

$$\eta = \frac{1}{2} \left(\int \int dx dp |W(x, p)| - 1 \right) \quad (5.1)$$

Here the normalization condition of the Wigner function was used to introduce 1 at the end. This definition can be easily extended to the discrete, finite-dimensional definition by replacing the integrals by sums over phase space points. Thus we have,

$$\eta = \frac{1}{2} \left(\sum_{\alpha} |W_{\alpha}| - 1 \right) \quad (5.2)$$

The quantity, η , has an upper bound that can be derived from its definition [18].

$$\eta \leq \frac{\sqrt{N} - 1}{2} \quad (5.3)$$

The central purpose of this study is to use the Wigner negativity as a probe of chaos in quantum systems. Such a study was conducted in [19] where the time evolution of the Wigner negativity was studied. The system used was a nonlinear kicked Kerr oscillator. They found that the negativity is a highly varying function of time for the parameters that give rise to chaos in the classical counterpart of the system. The paper also introduced an entropic measure of the negativity which showed smooth behaviour for integrable parameter regimes while it became highly non-monotonic in the regime where the classical system shows chaos.

Here the Wigner negativity of for quantum kicked rotor was studied. The negativity is a function of 5 parameters, N, α, β, κ , and time. The variation of the negativity was analyzed by varying each parameter while choosing a specific value of the other parameters. An important step in the numerical analysis was the choice of initial states. 3 different initial states were chosen.

Random state: the coefficients of this state expressed as $x + iy$ where each x and y was drawn from $\text{Normal}(0, 1)$

q-eigenstate: this is a slight misnomer. It is actually an eigenstate of the computational basis, i.e., the basis in which the Floquet operator was created numerically. In other words, only one coefficient is set to 1 while all others were set to 0.

Coherent state: the description was obtained from [15]. The coherent state on the torus can be obtained by first finding the ground state of the following hamiltonian:

$$\hat{H}_h = 2 - \frac{1}{2} \left(\hat{T}_q + \hat{T}_q^\dagger + \hat{T}_p + \hat{T}_p^\dagger \right)$$

This Hamiltonian is a quantization of the classical hamiltonian $2 - \cos(2\pi q) - \cos(2\pi p)$ on the torus. The eigenvalues of this hamiltonian and the hamiltonian described above in terms of the unitary operators are bounded between 0 and 4. Furthermore, expanding the classical hamiltonian around $q = p = 0$ to first order gives the harmonic oscillator hamiltonian. Thus we can assume that the ground state of the hamiltonian above will be similar to the ground state of the harmonic oscillator, i.e., it will be a minimum

uncertainty state. Denoting this state by $|00\rangle$ we can further define

$$|pq\rangle = \hat{T}_p^p \hat{T}_q^{-q} |00\rangle, \quad p, q \in [0, N-1]$$

The state $|pq\rangle$ is the required coherent state on the torus and it is centred at position and momentum values q and p respectively.

Using these initial states the variation of the negativity for each parameter was studied. For each of the parameters, except time, the initial state was evolved using the Floquet operator upto time $t \sim N$ which represents the Heisenberg time for this quantum standard map on the torus. Beyond this time scale the state begins resembling the initial state. In accordance with the quantum recurrence theorem [20] a time evolved state will begin resembling its initial state after a finite amount of time. Thus for each analysis, the time upto which the system is evolved is the Heisenberg time.

5.1 Time Evolution

The time evolution of the negativity show different behaviours in the integrable and chaotic regime [19]. For the kicked Kerr oscillator studied in [19], in the integrable case, the negativity increased to a maximum and slowly died out. However, for the chaotic case (i.e., the parameter values for which the classical system showed chaotic behaviour), the negativity showed extreme sensitivity at each kick. Analysis of the time evolution of negativity for the quantum kicked rotor would provide insights into how chaos manifests in the system studied here.

Before presenting the numerical results, some expectations about the behaviour can be inferred analytically. Given an initial state, $|\psi\rangle$ it can be expanded in the floquet eigenbasis as $\sum_k c_k |\phi_k\rangle$. The Wigner function as from eq. (3.10) can be written as

$$W_\alpha = \frac{1}{N} \text{Tr} \left(\hat{A}_\alpha \sum_{k,l} c_k c_l^* |\phi_k\rangle \langle \phi_l| \right)$$

The sum can be separated out into diagonal terms, i.e, $k = l$ and off-diagonal terms of the density matrix which leads to

$$W_\alpha = \frac{1}{N} \text{Tr} \left(\hat{A}_\alpha \sum_k |c_k|^2 |\phi_k\rangle \langle \phi_k| \right) + \frac{1}{N} \text{Tr} \left(\hat{A}_\alpha \sum_{k \neq l} c_k c_l^* |\phi_k\rangle \langle \phi_l| \right)$$

The first term in the series corresponds to a weighted sum of the Wigner functions of each Floquet

eigenstate. The second term is a weighted sum of the Wigner function arising from interference between different eigenstates of the Floquet operator. It is clear from the above relation that the time evolution of the Wigner function involves the 2nd term only. Thus, the behaviour of the 2nd term alone provides insights into how the negativity will vary over time for any generic initial state.

$$W_\alpha \xrightarrow{\hat{U}} W_\alpha = \frac{1}{N} \text{Tr} \left(\hat{A}_\alpha \sum_k |c_k|^2 |\phi_k\rangle \langle \phi_k| \right) + \frac{1}{N} \text{Tr} \left(\hat{A}_\alpha \sum_{k \neq l} c_k c_l^* e^{i(\theta_k - \theta_l)} |\phi_k\rangle \langle \phi_l| \right) \quad (5.4)$$

Each term can be rewritten as

$$W_{\alpha, \text{diag}} = \sum_k |c_k|^2 \left(\frac{1}{N} \text{Tr} \left(\hat{A}_\alpha |\phi_k\rangle \langle \phi_k| \right) \right) = \sum_k |c_k|^2 W_\alpha^{(kk)} \quad (5.5a)$$

$$W_{\alpha, \text{cross}} = \sum_{k \neq l} c_k c_l^* e^{i(\theta_k - \theta_l)} \left(\frac{1}{N} \text{Tr} \left(\hat{A}_\alpha |\phi_k\rangle \langle \phi_l| \right) \right) = \sum_k c_k c_l^* e^{i(\theta_k - \theta_l)} W_\alpha^{(kl)} \quad (5.5b)$$

There are 3 factors that influence the variation of negativity with time.

1. $\theta_k - \theta_l$: the qualitative behaviour is modulated by this phase.
2. c_k, c_l^* : the coefficients pick which Floquet eigenstates contribute to the initial state. Different initial states will have different pairs of k, l contributing.
3. Cross-Wigner function ($W_\alpha^{(kl)}$): this is created from the off-diagonal elements of the density matrix and shows the interference effects between different Floquet eigenstates. Depending on the regime it will have varying behaviour.

The variation of the negativity with time in the integrable regime for initial states chosen as a coherent state and a q-eigenstate were oscillatory in nature. For the first few kicks the range of oscillations was large. The range reduced from ~ 15 kicks onwards. Beyond that point the value oscillated within a small range. The q-eigenstate showed similar behaviour qualitatively. In the chaotic regime, the negativity of the coherent state increased to around its maximum within on the second kick. From the second kick onwards, the negativity was more or less constant within a range of approximately 0.05. The q-eigenstate, jumped to its maximum value on the first kick and like the coherent state, oscillated around the maximum within a range of 0.05. The behaviour of the random state did not show any such patterns in either regime. For both, the integrable and chaotic regimes, the negativity for the random state oscillated in a narrow range without any visible patterns. Fig 5.1 shows the time evolution in the integrable regime for the coherent state and the q-eigenstate.

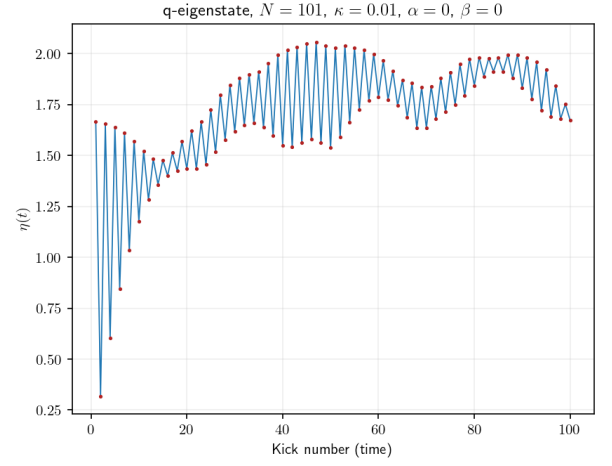
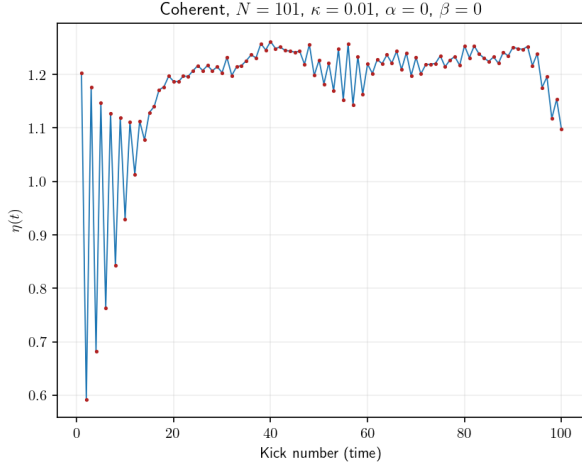
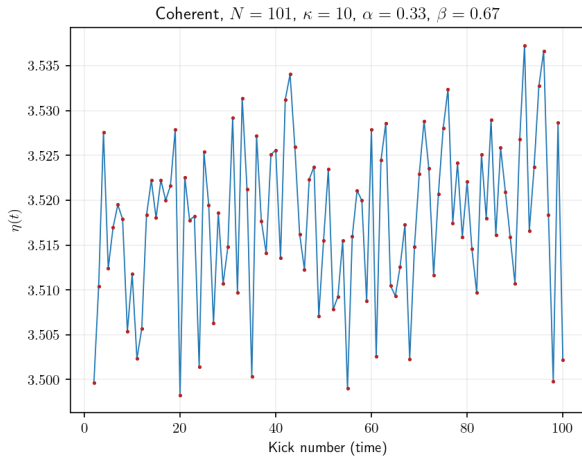
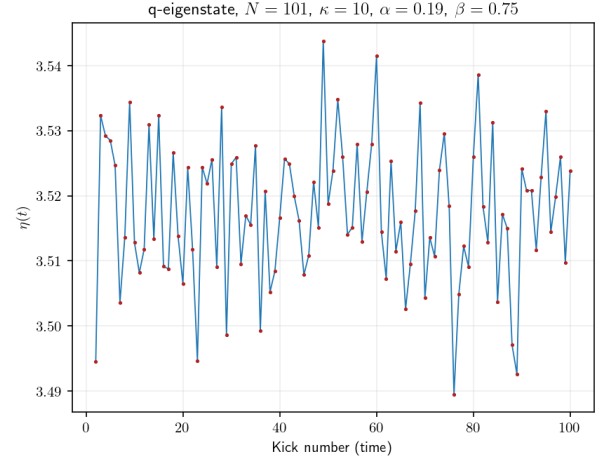


Figure 5.1: Time evolution of the negativity in the integrable regime for the coherent state and q-eigenstate. Parameters: $N = 101$, $\kappa = 0.01$, $\alpha = \beta = 0$. The 0-th time step is not shown.

Fig. 5.2 shows the negativity variation in the chaotic regime for the coherent state and the q-eigenstate.



(a)



(b)

Figure 5.2: Time evolution of the negativity in the chaotic regime for the coherent state and q-eigenstate. Parameters: $N = 101$, $\kappa = 10$, $\alpha \neq \beta \neq 0, 0.5$. The variation from the second kick onwards is shown to emphasize the absence of any pattern in the variation.

It is evident from Fig. 5.2 that there is not discernable pattern in the negativity with time. Furthermore, it randomly oscillates within a small range which is the same for both. The oscillations are not pure, i.e., the direction of the change in the value of negativity does not alternate between kicks. It increases for 2-3 kicks at some points but at some it increases at one kick and decreases right after.

The behaviour for each initial state is determined by the 3 factors mentioned above. Its qualitative behaviour largely depends on the phase difference that is introduced at each kick. The circulant versus non-circulant property of the Floquet operator is the major contributing factor to the presence of period-

icity in η with each kick. In the integrable regime we have $\kappa \approx 0$ and $\beta = 0$. In such a case the Floquet operator is exactly circulant, i.e., the columns are cyclic permutations of each other. The matrix can thus be represented as

$$C = \begin{pmatrix} c_0 & c_{N-1} & \cdots & c_2 & c_1 \\ c_1 & c_0 & c_{N-1} & \cdots & c_2 \\ \vdots & c_1 & c_0 & \ddots & \vdots \\ c_{N-2} & \vdots & \ddots & \ddots & c_{N-1} \\ c_{N-1} & c_{N-2} & \cdots & c_1 & c_0 \end{pmatrix}$$

While this shows that the diagonal element does not change at all, it is possible in the general case that the diagonal element is different but it will be one of some from the first column. For a circulant matrix the eigenvectors are the Fourier modes,

$$v_j = \frac{1}{\sqrt{N}}(1, \omega^j, \omega^{2j}, \dots, \omega^{n-1}j)^T$$

Where $\omega = e^{(2i\pi/N)}$ is a primitive root of unity. Thus, the coefficients of the Floquet eigenstates are permutations of the ordered set of the j -th roots of unity. The eigenvalues are linear combinations of the conjugated roots of unity. Thus the difference $\theta_k - \theta_l$ involves a linear combination of roots of unity which induce oscillations at each kick. The magnitude of the oscillations are modulated by the factor $c_k c_l^*$. The cross-Wigner function in this case has a plane wave form since the outer product $|\phi_k\rangle\langle\phi_l|$ has elements which are products and sums of the roots of unity. Thus the whole time evolution of the negativity involves a sum of complex exponentials which are modulated by the factor $c_k c_l^*$. Thus, in the integrable regime there are oscillations of the negativity observed.

However, in the chaotic case both κ and β are non-zero. Both of these factors break the circulant structure of $\hat{U}$ which means that the Floquet eigenstates lose the dependence on roots of unity. The eigenstates become spread out in phase space and behave like random vectors with no structure. As a result of this the oscillations observed in the negativity are destroyed. Furthermore, since the states are essentially like random vectors, their Wigner function are very spread out. As a result, there are almost equal parts of positive and negative volumes in the Wigner function for chaotic Floquet eigenstates. This leads to a sort of averaging of contributions to the negativity. These two reasons in conjunction with each other lead to the plots seen in figs. 5.2 where there is random evolution and the range is very small.

The random state, by virtue of being random, behaves similarly in each regime. It varies over a very

small range. In the integrable regime there is a semblance of periodicity while in the chaotic regime there is no discernable pattern. The spectral decomposition in either regime confirms the obvious; almost all of the Floquet eigenstates overlap upto varying degrees with the random state. Fig. 5.3 shows the time evolution of the random state with time in the integrable and the chaotic regime. The appearance of slight

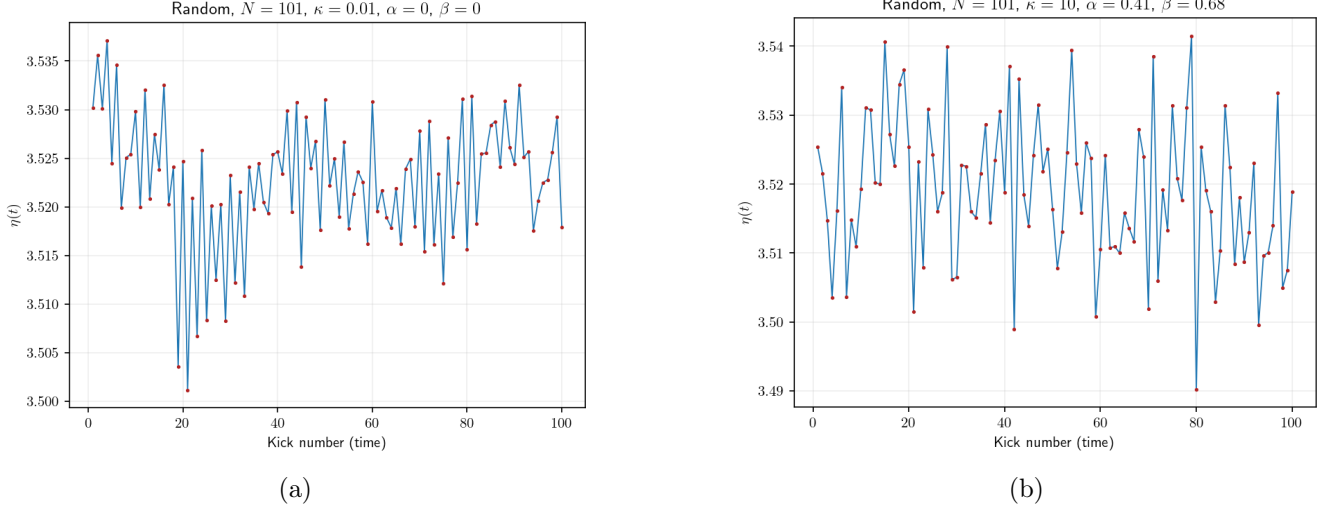


Figure 5.3: Time evolution of negativity of the random state for $N = 101$, and (a) $\kappa = 0.01$, $\alpha = \beta = 0$, (b) $\kappa = 10$, $\alpha \neq \beta \neq 0, 0.5$

periodicity in fig. 5.3a can be attributed to the circulant structure of $\hat{U}$. In the chaotic regime the absence of such a property leads to the same behaviour as for the other 2 states.

5.2 Coupling Constant

In the classical system the coupling constant is the only definitive parameter which determines if the system is integrable or chaotic. In the classical system $\kappa \sim 5$ is where invariant tori encircling the entire phase space begin breaking up and at around $\kappa \sim 10$, the system becomes completely chaotic. There is a value $\kappa_c \sim 0.9716$ which marks a transition from integrability to non-integrability. This value of the coupling constant is where the trajectories winding around the phase space in the q direction break up allowing for the mixing of phase space [21]. The dependence of the negativity on the coupling constant is useful to understand the links between the classical and quantum system since negativity is a quantum feature while the coupling constant determines the kind of dynamics in the classical case.

Fig. 5.4 shows the variation of the negativity with coupling constant. The negativity seems to be extremely sensitive to the coupling constant. For the random state there seems to be no dependence at all. The q-eigenstate is different from the other two since it stabilizes at around $\kappa = 1$. The coherent state negativity

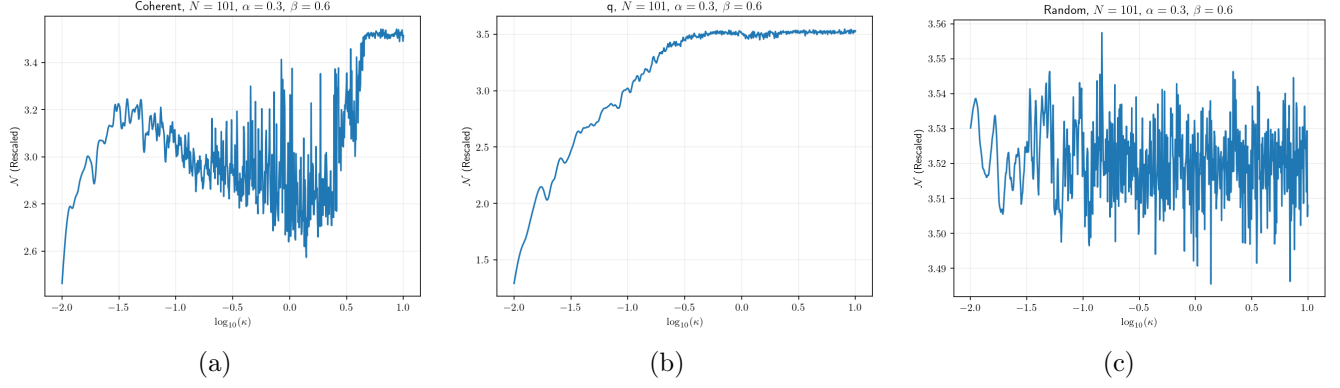


Figure 5.4: Variation of the negativity with coupling constant for the 3 initial states for $N = 101$, $\alpha = 0.3$, $\beta = 0.6$

also becomes constant from around $\kappa = 3.2$. These plots do not provide much information since there seems to be no discernable pattern. It thus remains open to further analysis in the future.

5.3 Symmetry Parameters

The parameters α and β induce quasi-periodic boundary conditions on the torus phase space and control parity and time reversal respectively. The classical system does not possess such boundary condition. Being a purely quantum variable, probing the variation of η with α and β would provide a glimpse into the quantum nature of the chaos of the kicked rotor.

Numerical calculations show interesting dependencies of η on the symmetry parameters. The behaviour is markedly different in the integrable and the chaotic regimes.

Numerical calculations show interesting dependencies of η on the symmetry parameters. η was visualized in an (α, β) “phase space” (referred to as (α, β) space hereafter). The results showed dependence on the value of N and κ reflecting a transition in behaviour. The negativity was calculated for multiple values of α and β ranging from 0 to 1 while choosing κ to be 0.01 and 10 where each represents the integrable and maximally chaotic state of the classical system. It was noted that the range over which η varied changed as a function of N and κ . For $\kappa = 0.01$, all values of N show no variation of η in α for a fixed β while for the opposite case, i.e., varying β and fixed α , there was a definite periodic variation. Thus, in the integrable regime there is a dependence on N of how much the variation of α contributes to η while the β dependence does not vanish with increasing N .

In the maximally chaotic case, the behaviour was independent of N . Regardless of what the value of N was, for $\kappa = 10$, the range over which η varied for either α or β reduced. The (α, β) space became very

mixed for any value of N in the chaotic regime.

The difference in the variation can be visualized through the (α, β) phase plots and the corresponding slices at specific values of each parameter. Figs. 5.5a and 5.5b show the phase plot for a particular set of parameter values. In the integrable regime the dependence on β is much more than α . In the chaotic regime the effect of β gets reduced.

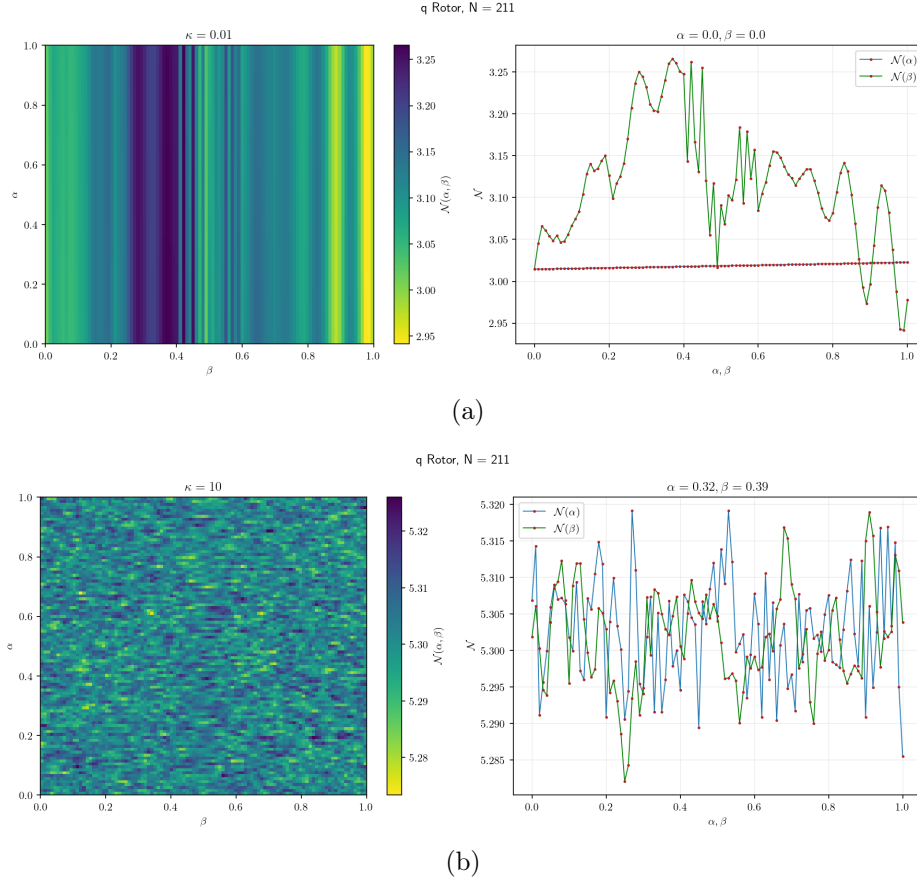


Figure 5.5: Phase plot of $\eta(\alpha, \beta)$ for (a) $N = 211, \kappa = 0.01, \alpha = \beta = 0$ and (b) $N = 211, \kappa = 10, \alpha \neq \beta \neq 0, 0.5$

Beyond a visual analysis it is informative to see how the range over which η varies for either α or β itself varies as a function of system size. In the integrable regime

For initial states chosen to be the coherent state and the q-eigenstate, it is evident that the dependence of η on α reduces with increasing N but increases on β .

The results obtained can be explained by the role played by α and β in the matrix elements of the Floquet operator in the position basis. Before that it is instructive to note how the elements of the density matrix

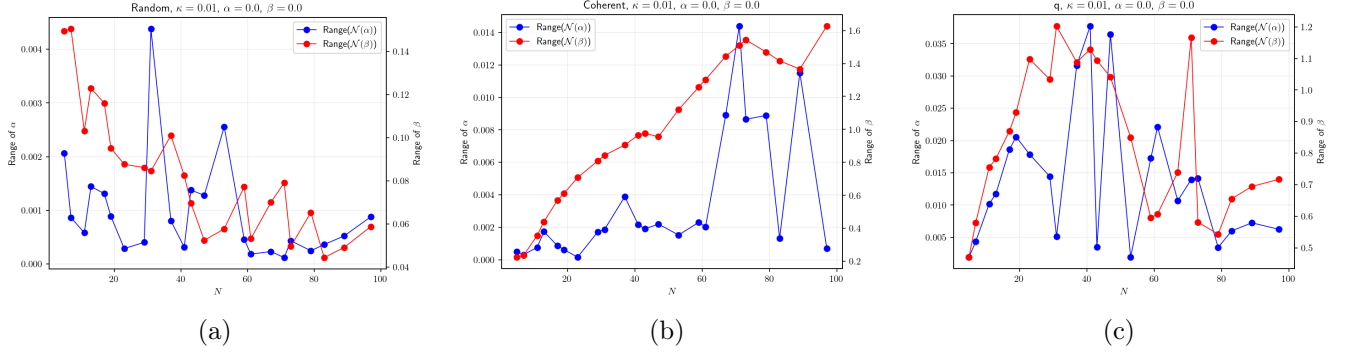


Figure 5.6: Range of variation of negativity with α and β are plotted separately to understand the dependence on the role played by α and β as a function of system size.

evolve under time evolution. A single kick changes the density matrix as

$$\hat{\rho}' = \hat{U} \hat{\rho} \hat{U}^\dagger$$

Introducing resolutions of identity in terms of the position basis in between gives

$$\begin{aligned} \hat{\rho}' &= \sum_{j,k} \hat{U} |j\rangle \langle j| \hat{\rho} |k\rangle \langle k| \hat{U}^\dagger \\ \Rightarrow \hat{\rho}' &= \sum_{j,k} \rho_{jk} \hat{U} |j\rangle \langle k| \hat{U}^\dagger \end{aligned}$$

Introducing $|m\rangle$ on the right and $\langle n|$ on the left to obtain the matrix elements of $\hat{\rho}'$

$$\rho'_{mn} = \sum_{j,k} U_{nj} U_{mk}^* \rho_{jk}$$

Now consider the form of the Floquet operator given in (4.10). There are two distinct parts to it, one which arises from the kick term, where α appears and the other from the momentum term of the Hamiltonian, where β appears. To refresh memory, recall the kick and momentum terms. Denoting the operator elements as $U_{nn'} = (1/N) \mathcal{K}(n, \alpha) \mathcal{F}(n - n', \beta)$ (the dependence on N and κ is implicit), we first consider the kick term,

$$\mathcal{K}(n, \alpha) = \exp \left(\frac{iN\kappa}{2\pi} \cos \left(\frac{2\pi}{N} (n + \alpha) \right) \right)$$

Given that α appears only in the argument of the cosine in the α/N form we can deduce that the ratio $\alpha/N \ll 1$ for a large value of N since $\alpha \in [0, 1)$. Thus adding a small value to the argument of a sinusoidal function does not change its value much which shows that for large values of N , the matrix elements of

the Floquet operator do not get appreciably affected by α .

The dependence on β is slightly more subtle. The function $\mathcal{F}(m, n - n', \beta)$ is

$$\mathcal{F}(n - n', \beta) = \sum_{m=0}^{N-1} \exp\left(-\frac{i\pi}{N}(m + \beta)^2\right) \exp\left(\frac{2i\pi}{N}(m + \beta)(n - n')\right)$$

From here onwards, $k = n - n'$. The exponent can be simplified to extract the terms without m appearing

$$-\frac{i\pi}{N}(m + \beta)^2 + \frac{2i\pi}{N}k(m + \beta) = \frac{i\pi}{N}(2k(m + \beta) - (m + \beta)^2) = \frac{i\pi}{N}(2k\beta - \beta^2 + 2m(k - \beta) - m^2)$$

Thus the function $\mathcal{F}(k, \beta)$ becomes

$$\mathcal{F}(m, k, \beta) = e^{\frac{i\pi}{N}(2k\beta - \beta^2)} \sum_{m=0}^{N-1} e^{\frac{i\pi}{N}(2m(k - \beta) - m^2)}$$

The summation would be an exact Gaussian quadratic sum for $\beta = 0$ which has known values. Since β is not an integer, the summation is not an exact Gaussian sum. Nevertheless, it can be considered to be a complex number which can be expressed in polar form as $Ce^{i\theta(k, \beta)}$ where θ is the argument of the complex number and it is dependent on k and β . With this the function has the form

$$\mathcal{F}(k, \beta) = e^{-\frac{i\pi\beta^2}{N}} C e^{\frac{i\pi}{N}(2k\beta + N\theta(k, \beta))}$$

From the functional forms of the different components of the matrix elements it is evident that there are two kinds of phase factors dependent on β . The factor $e^{-i\pi\beta^2/N}$ is a global phase factor which will get cancelled due to complex conjugation when the Floquet operator acts on the density matrix. This is evident from the equation that relates the time evolved density matrix elements to the original density matrix elements derived above. However, the term that depends on $k = n - n'$ introduces a local phase which has an effect on the elements of the density matrix at each kick. This local phase is also dependent on β . As a result the evolution of the states bears a mark of β which consequently affects the value of the negativity.

The explanation provided above holds in the integrable regime only. In the chaotic regime the product $N\kappa$ becomes appreciable and leads to a suppression of the contribution from β as well. To understand this consider the product of the exponentials in $\mathcal{K}(n, \alpha)$ and $\mathcal{F}(k, \beta)$ which leads to a summation of the exponents. The local phase, i.e., the terms dependent on n and n' only are (not writing the factor inside

the cos for easier notation)

$$\text{Local phase} = \frac{iN\kappa}{2\pi} \cos(\dots) + \frac{2k\beta\pi}{N} + \pi\theta(k, \beta)$$

The term $\theta(k, \beta)$ is the argument of a complex number written in polar form so it will always be in $[0, 2\pi]$. The factor $k/N \in [-1, 1]$ since $k \in [-(N-1), (N-1)]$. In the chaotic regime $N\kappa$ is not a fraction and is an appreciable number and even for $N = 5$ it is 50 which introduces a big factor to each matrix element and dominates over β for most of the elements of the Floquet operator. Thus in the chaotic regime the dependence of β is also washed out.

5.4 System Size

Negativity as a function of system size is known to increase. Since the upper bound of η is $(\sqrt{N} - 1)/2$ it is worthwhile to track the growth of negativity rescaled by $\sqrt{N}$.

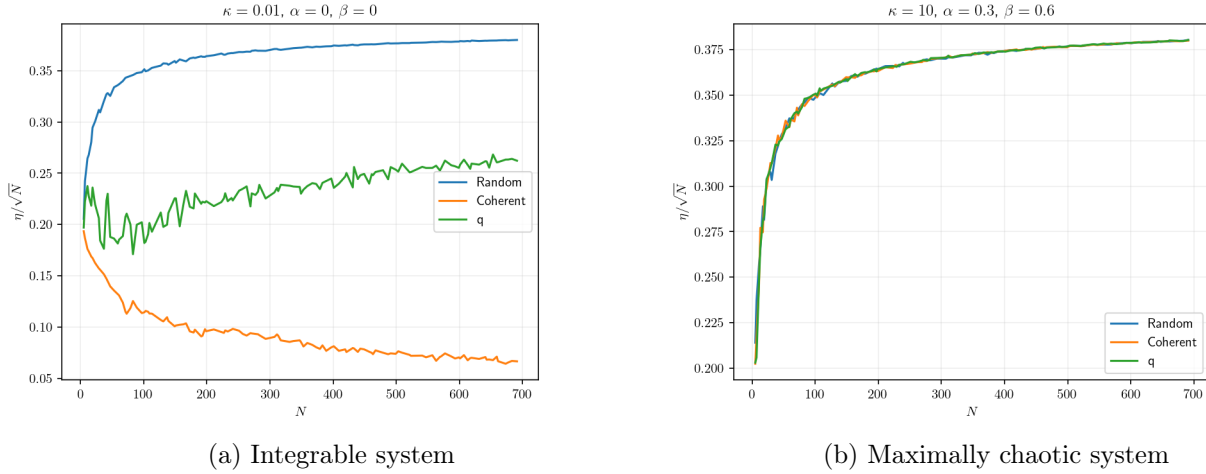


Figure 5.7: Variation of η with system size.

The variation of the negativity becomes of $\mathcal{O}(\sqrt{N})$ for all cases for larger system sizes. In the integrable case the value at which it saturates is different. However, for the chaotic case regardless of the initial state the growth of negativity is the same with N .

Chapter 6

Conclusion

The Wigner function provides a quasi-phase space structure which allows the understanding of quantum chaos in a different way. Negativity of the Wigner function, which serves as a measure of non-classicality, also provides insights into how quantum chaos ensues. This was studied for the quantum kicked rotor here. The most detailed signatures arise from the time evolution. In the integrable regime the Floquet operator is a circulant matrix which leads to the oscillations seen for the coherent state and the q-eigenstate. The introduction of a non κ breaks that structure and causes the reduction in the range of variation while removing the oscillatory character of η . The current analysis has established a correlation between the structure of the Floquet operator and the time evolution of η . A natural extension of this is to understand, in more depth, the exact contribution of the circulant structure to the time evolution. An analysis of the overlap coefficients and their contributions would also be of help. In addition to that visualizing and getting a sense of the cross-Wigner functions would provide additional help in understanding the reason why the time evolution behaves the way it does.

The analysis of the variation of η with each parameter has shown that it is quite difficult to separate the dependencies. Choosing different fixed values shows slightly different behaviour which is to be expected. One of the most striking features is shown by the system size where the scaling in the chaotic regime is independent of the initial state. This provides a different perspective to realize that the states, in the chaotic regime, get scrambled almost instantaneously due to the nature of $\hat{U}$. The analysis of the dependence on the coupling constant remains incomplete. The fact the time evolution of the negativity is dependent on the regime is why the variation with κ is hard to analyze since the negativity of the state evolved to Heisenberg time is used to plot the variation with system parameters. Furthermore, the absence of a sharp transition from integrability to chaos prevents choosing the point upto which β remains 0 for

the variation with the coupling constant. The question of the dependence on κ remains open.

This thesis looked at the quantum kicked rotor on a torus only. A natural follow up is to study the same thing for different quantum maps such as the Harper map, Baker's map, and so on. Analyzing the negativity for different systems is crucial in separating model dependent and model independent behaviour. Since the overarching aim is to understand quantum chaos in a different way and possibly provide a more rigorous metric, the presence of model independent features of the negativity would provide a good direction.

Appendix

Proof: Average of observable eq. (2.5)

The average of an observable in terms of the density matrix is given by

$$\langle A \rangle = \text{Tr}(\hat{\rho}\hat{A})$$

Given the form of the Wigner function and the inverse-Weyl transform we can write

$$\int \int dx dp A(x, p) W(x, p) = \frac{1}{2\pi\hbar} \int dx dp dy dz e^{-ipy/\hbar} \langle x + y/2 | \hat{\rho} | x - y/2 \rangle e^{ipz/\hbar} \langle x - z/2 | \hat{A} | x + z/2 \rangle$$

The integral over p gives $\delta(z - y)$ which then if we integrate over z will give us

$$\langle A \rangle = \int dx dy \langle x + y/2 | \hat{\rho} | x - y/2 \rangle \langle x - y/2 | \hat{A} | x + y/2 \rangle$$

Changing variables as $u = x + y/2$ and $v = x - y/2$ we get

$$\langle A \rangle = \int du dv \langle u | \hat{\rho} | v \rangle \langle v | \hat{A} | u \rangle = \int du \langle u | \hat{\rho} \hat{A} | u \rangle = \text{Tr}(\hat{\rho}\hat{A})$$

Proof: Inverse Weyl Transform eq. (2.7)

Given that the Weyl transform is the correct mapping from a phase space function to the operator the inverse transform can be derived from it. It will be done in the position representation. We look for the matrix elements of $\langle y | \hat{A} | z \rangle$ where $|y\rangle$ and $|z\rangle$ are position basis kets.

Furthermore the operator $e^{i(\sigma\hat{x}+\tau\hat{p})/\hbar}$ can be decomposed using the Baker-Campbell-Hausdorff formula which states $e^{\hat{A}+\hat{B}} = e^{\hat{A}}e^{\hat{B}}e^{-[A,B]/2}$. The commutator in our case is $[i\sigma\hat{x}/\hbar, i\tau\hat{p}/\hbar] = -i\sigma\tau/\hbar$. Thus the operator can be decomposed as $e^{i(\sigma\hat{x}+\tau\hat{p})/\hbar} = e^{i\sigma\hat{x}/\hbar}e^{i\tau\hat{p}/\hbar}e^{i\sigma\tau/2\hbar}$. In (2.6) we look for the matrix elements

in the position representation, $\langle y|\hat{A}|z\rangle$.

$$\begin{aligned}\langle y|\hat{A}|z\rangle &= \frac{1}{(2\pi\hbar)^2} \int d\sigma d\tau dx dp A(x, p) e^{-i(\sigma x + \tau p + \sigma\tau/2)/\hbar} \langle y|e^{i\sigma\hat{x}/\hbar} e^{i\tau\hat{p}/\hbar}|z\rangle \\ &= \frac{1}{(2\pi\hbar)^2} \int d\sigma d\tau dx dp A(x, p) e^{-i(\sigma x + \tau p + \sigma\tau/2)/\hbar} e^{i\sigma z/\hbar} \langle y|z + \tau\rangle\end{aligned}$$

Performing the integral over σ would give $2\pi\hbar\delta(z - x + \tau/2)$. After that doing the integral over x would convert $A(x, p) \rightarrow A(z + \tau/2, p)$ and we also have $\langle y|z + \tau\rangle = \delta(y - z - \tau)$. With all of this we get

$$\begin{aligned}\langle y|\hat{A}|z\rangle &= \frac{1}{2\pi\hbar} \int d\tau dp A(z + \tau/2, p) e^{-i\tau p/\hbar} \delta(y - z - \tau) \\ &= \frac{1}{2\pi\hbar} \int dp A\left(\frac{z+y}{2}, p\right) e^{-i(z-y)p/\hbar}\end{aligned}$$

Changing variables to $u = \frac{z+y}{2}$, $v = z - y$. Then $z = u + v/2$ and $y = u - v/2$. Thus we get

$$\langle u - v/2|\hat{A}|u + v/2\rangle = \frac{1}{2\pi\hbar} \int dp A(u, p) e^{-ivp/\hbar}$$

Multiplying on both sides with $e^{iv'p/\hbar}$ and integrating over dv' we get

$$\begin{aligned}\int dv' \langle u - v/2|\hat{A}|u + v/2\rangle e^{ipv'/\hbar} &= \frac{1}{2\pi\hbar} \int dp dv' A(u, p) e^{i(v'-v)p/\hbar} \\ \Rightarrow \int dv' \langle u - v/2|\hat{A}|u + v/2\rangle e^{ipv'/\hbar} &= A(u, p) \delta(v' - v)\end{aligned}$$

The equation will be non-trivial only if $v' = v$. Furthermore we can just replace u with x for notational consistency and we get (2.7)

$$A(x, p) = \int dv \langle u - v/2|\hat{A}|u + v/2\rangle e^{ipv'/\hbar}$$

Proof: *Time Evolution eq. (2.14)*

The full derivation is given in the Appendix. Taking a time derivative of the Wigner function in the pure state form gives

$$\frac{\partial W}{\partial t} = \frac{1}{2\pi\hbar} \int dy e^{ipy/\hbar} \left[\psi^*(x + y/2) \frac{\partial \psi(x - y/2)}{\partial t} + \psi(x - y/2) \frac{\partial \psi^*(x + y/2)}{\partial t} \right]$$

Replacing the terms from Schrödinger's equation we can also divide the two time dependent terms into a

kinetic and potential part such that

$$\frac{\partial W}{\partial t} = \frac{\partial W_K}{\partial t} + \frac{\partial W_V}{\partial t}$$

The kinetic part simply is

$$\frac{\partial W_K}{\partial t} = \frac{i}{4\pi m} \int dy e^{ipy/\hbar} \left[\psi^*(x+y/2) \frac{\partial^2 \psi(x-y/2)}{\partial x^2} - \psi(x-y/2) \frac{\partial^2 \psi^*(x+y/2)}{\partial x^2} \right]$$

The second derivative in terms of x can be converted to a mixed partial derivative in x and y by noting that $\frac{\partial^2}{\partial x^2} = \pm 2 \frac{\partial^2}{\partial x \partial y}$. As a result the above equation becomes:

$$\frac{\partial W_K}{\partial t} = -\frac{i}{2\pi m} \int dy e^{ipy/\hbar} \left[\psi^*(x+y/2) \frac{\partial^2 \psi(x-y/2)}{\partial x \partial y} + \psi(x-y/2) \frac{\partial^2 \psi^*(x+y/2)}{\partial x \partial y} \right]$$

This term now becomes easier to integrate by parts. On integrating by parts the first boundary term will go to 0. The second term on integrating by parts for the first term on the right of the above equation becomes:

$$-\frac{i}{2\pi m} \int dy e^{ipy/\hbar} \psi^*(x+y/2) \frac{\partial^2 \psi(x-y/2)}{\partial x \partial y} = \frac{i}{2\pi m} \int dy \frac{\partial \psi(x-y/2)}{\partial x} e^{ipy/\hbar} \left\{ \frac{ip}{\hbar} \psi^*(x+y/2) + \frac{1}{2} \frac{\partial \psi^*(x+y/2)}{\partial y} \right\}$$

Similarly the 2nd term on the right becomes

$$-\frac{i}{2\pi m} \int dy e^{ipy/\hbar} \psi(x-y/2) \frac{\partial^2 \psi^*(x+y/2)}{\partial x \partial y} = \frac{i}{2\pi m} \int dy \frac{\partial \psi(x-y/2)}{\partial x} e^{ipy/\hbar} \left\{ \frac{ip}{\hbar} \psi(x-y/2) - \frac{1}{2} \frac{\partial \psi(x-y/2)}{\partial y} \right\}$$

Adding these two terms will get rid of the terms with the factor $1/2$ in front of them if we note that $\frac{\partial}{\partial x} = \frac{\partial}{\partial y}$.

Thus we have

$$\frac{\partial W_K}{\partial t} = -\frac{p}{m} \frac{1}{2\pi\hbar} \int dy e^{ipy/\hbar} \left[\psi(x-y/2) \frac{\partial \psi^*(x+y/2)}{\partial x} + \psi^*(x+y/2) \frac{\partial \psi(x-y/2)}{\partial x} \right] = -\frac{p}{m} \frac{\partial W}{\partial x}$$

The solution of the potential term assumes that the potential can be Taylor expanded into a series. Thus the difference can be written as

$$V(x+y/2) - V(x-y/2) = \sum_{n=0}^{\infty} \frac{1}{n!} \frac{\partial^n V(x)}{\partial x^n} \left(\left(\frac{y}{2} \right)^n + \left(-\frac{y}{2} \right)^n \right) = \sum_{s=0}^{\infty} \frac{1}{(2s+1)!} \left(\frac{\partial^{2s+1}}{\partial x^{2s+1}} \right) V(x) \frac{y^{2s+1}}{2^{2s}}$$

Plugging this into the integral and placing the summation outside the integral sign we get

$$\begin{aligned}
\frac{\partial W_V}{\partial t} &= \frac{i}{2\pi\hbar^2} \sum_{s=0}^{\infty} \frac{1}{(2s+1)!} \frac{1}{2^{2s}} \left(\frac{\partial^{2s+1}}{\partial x^{2s+1}} \right) V(x) \int dy e^{ipy/\hbar} y^{2s+1} \psi^*(x+y/2) \psi(x-y/2) \\
&= \frac{i}{\hbar} \sum_{s=0}^{\infty} \frac{1}{(2s+1)!} \frac{1}{2^{2s}} \left(\frac{\partial^{2s+1}}{\partial x^{2s+1}} \right) V(x) \left(\frac{\hbar}{i} \right)^{2s+1} \frac{\partial^{2s+1}}{\partial p^{2s+1}} \left(\frac{1}{2\pi\hbar} \int dy e^{ipy/\hbar} \psi^*(x+y/2) \psi(x-y/2) \right) \\
&= \sum_{s=0}^{\infty} \frac{1}{(2s+1)!} \left(\frac{\hbar}{2i} \right)^{2s} \left(\frac{\partial^{2s+1}}{\partial x^{2s+1}} \right) V(x) \left(\frac{\partial^{2s+1}}{\partial p^{2s+1}} \right) W(x, p)
\end{aligned}$$

Thus finally the time evolution of the Wigner function is

$$\frac{\partial W}{\partial t} = -\frac{p}{m} \frac{\partial W(x, p)}{\partial x} + \sum_{s=0}^{\infty} \frac{1}{(2s+1)!} \left(\frac{\hbar}{2i} \right)^{2s} \left(\frac{\partial^{2s+1}}{\partial x^{2s+1}} \right) V(x) \left(\frac{\partial^{2s+1}}{\partial p^{2s+1}} \right) W(x, p) \quad (6.1)$$

Proof: *Discrete Evolution Equation eq. (3.21)*

Begin from the von-Neumann equation:

$$\frac{d\hat{\rho}}{dt} = -\frac{i}{\hbar} [\hat{H}, \hat{\rho}]$$

The discrete Weyl transform for the Hamiltonian is $\hat{H} = \sum_{\alpha} H_{\alpha} \hat{A}_{\alpha}$. The equation for the density matrix is $\hat{\rho} = \sum_{\alpha} W_{\alpha} \hat{A}_{\alpha}$. Using this the LHS is equivalent to

$$\frac{d(\sum_{\alpha} W_{\alpha} \hat{A}_{\alpha})}{dt} = \sum_{\alpha} \left(\frac{dW_{\alpha}}{dt} \hat{A}_{\alpha} \right)$$

The RHS becomes

$$\begin{aligned}
-\frac{i}{\hbar} [\hat{H}, \hat{\rho}] &= -\frac{i}{\hbar} \left[\sum_{\beta} H_{\beta} \hat{A}_{\beta}, \sum_{\gamma} W_{\gamma} \hat{A}_{\gamma} \right] \\
&= -\frac{i}{\hbar} \sum_{\beta, \gamma} H_{\beta} W_{\gamma} [\hat{A}_{\beta}, \hat{A}_{\gamma}] \\
&= -\frac{i}{\hbar} \sum_{\beta, \gamma} H_{\beta} W_{\gamma} (\hat{A}_{\beta} \hat{A}_{\gamma} - \hat{A}_{\gamma} \hat{A}_{\beta})
\end{aligned}$$

On both sides now we can multiply by $\hat{A}_{\alpha}$ and take a trace over the products. The LHS simply becomes

$N \frac{dW_\alpha}{dt}$. The RHS is

$$\begin{aligned} -\frac{i}{\hbar} \text{Tr} \left(\sum_{\beta, \gamma} H_\beta W_\gamma (\hat{A}_\beta \hat{A}_\gamma - \hat{A}_\gamma \hat{A}_\beta) \right) &= -\frac{i}{\hbar} \sum_{\beta, \gamma} H_\beta W_\gamma (\text{Tr}(\hat{A}_\beta \hat{A}_\gamma \hat{A}_\alpha) - \text{Tr}(\hat{A}_\gamma \hat{A}_\beta \hat{A}_\alpha)) \\ &= N \sum_{\beta, \gamma} H_\beta W_\gamma (\Gamma_{\beta\gamma\alpha} - \Gamma_{\gamma\beta\alpha}) \end{aligned}$$

Using the cyclicity of the trace we have $\Gamma_{\alpha\beta\gamma} = \Gamma_{\beta\gamma\alpha} = \Gamma_{\gamma\alpha\beta}$. Consider the second term in the sum $\sum_{\beta, \gamma} H_\beta W_\gamma \Gamma_{\gamma\beta\alpha}$. In this the indices label the summations. The indices can just be relabelled and swapped without any change in the summation. Thus the term becomes $\sum_{\beta, \gamma} W_\beta H_\gamma \Gamma_{\beta\gamma\alpha}$. Using the cyclicity of the trace and cancelling the N on both sides we recover the discrete evolution equation

$$\frac{dW_\alpha}{dt} = -\frac{i}{\hbar} \sum_{\beta, \gamma} \Gamma_{\alpha\beta\gamma} (H_\beta W_\gamma - W_\beta H_\gamma)$$

Proof: *Time evolution operator of periodically kicked Hamiltonian eq. (4.5)*

Consider the state of a system right before and right after the kick to be labelled by a superscript $-$ or $+$ respectively. In that case the problem of solving the evolution of a system defined by eq. (4.5) is to find the operator $\hat{U}$ such that

$$|\psi_{n+1}^+\rangle = \hat{U} |\psi_n^+\rangle$$

Given the Hamiltonian in eq. (4.5) Schrödinger's equation takes the form:

$$i\hbar \frac{d|\psi\rangle}{dt} = \hat{H}_0 |\psi\rangle + \delta_T \hat{V} |\psi\rangle$$

Where $\delta_T = \sum_{n=-\infty}^{\infty} \delta(t/T - n)$

Between two kicks, i.e., right after one and right before another, the kick operator does not affect the system. Thus the evolution is

$$|\psi_{n+1}^-\rangle = \exp(-i\hat{H}_0 T/\hbar) |\psi_n^+\rangle$$

On the other hand right before and right after a kick the system is affected by the time dependent kick

operator. Thus Schrödinger's equation with $\hat{H}_0$ effectively zero requires us to solve

$$i\hbar \frac{d|\psi\rangle}{dt} = \delta_T \hat{V} |\psi\rangle$$

$$|\psi_{n+1}^+\rangle = \exp\left(-i \frac{\hat{V}}{\hbar} \int_{t'}^t \delta_T(\tau) d\tau\right) |\psi_{n+1}^-\rangle$$

Integrating this over one period reduces to

$$|\psi_{n+1}^+\rangle = \exp\left(-i \hat{V} T / \hbar\right) |\psi_{n+1}^-\rangle$$

Putting everything together finally gives eq. (4.8)

$$|\psi_{n+1}\rangle = \exp\left(-i \hat{V} T / \hbar\right) \exp\left(-i \hat{H}_0 T / \hbar\right) |\psi_n\rangle$$

Proof: Upper bound of $\mathcal{N}$ (5.3)

Proof is also given in [18]. To obtain the upper bound of $\mathcal{N}$ we note that

$$\sum_{\alpha} W_{\alpha}^2 = \frac{1}{N} \text{Tr}(\hat{\rho}^2)$$

Application of Cauchy-Schwarz inequality provides a relation for the L1-norm in the N^2 dimensional space spanned by the phase point operators. The sum $\sum_{\alpha} |W_{\alpha}|$ is the L1 norm.

$$\begin{aligned} \sum_{\alpha} |W_{\alpha}| &= \sum_{\alpha} |W_{\alpha} \times 1| \\ &\leq (\sum_{\alpha} W_{\alpha}^2)^{1/2} \cdot (\sum_{\alpha} 1)^{1/2} \\ &= \sqrt{N} \text{Tr}(\hat{\rho}^2) \end{aligned}$$

For any generic state, $\text{Tr}(\hat{\rho}^2) \leq 1$ thus giving

$$\sum_{\alpha} |W_{\alpha}| \leq \sqrt{N}$$

Placing this in the definition of $\mathcal{N}$ provides the bound.

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